

The Causal Effects of Korea's Quota Tariff Policy on Agricultural Product Retail Prices

Deokjae Jeong Youngmi Kim

Korea Customs and Trade Development Institute

Abstract

The Korean government operates the Quota Tariff system with the objective of stabilizing prices. This study examines whether the policy causally lowered agricultural retail prices as intended. Quota Tariff designations begin at different times across products (staggered adoption), the treatment is suspended and later resumed (non-absorbing), and the depth of the tariff reduction also differs across products (heterogeneous treatment intensity). Since existing estimation methods that presuppose absorbing treatment cannot be applied to this structure, we adopt local projection difference-in-differences as the best-suited method. We then estimate two effects: the effect while the tariff reduction remains in force (the persistent-treatment specification) and the total effect of initiating a Quota Tariff designation (the impulse-response specification). For the vegetable group, retail price declines were insignificant in both specifications, whereas the fruit group showed declines of about 0.78 percent per one percentage point of tariff reduction at 150 days after Quota Tariff onset (persistent treatment) and 0.49 to 0.63 percent at about 250–273 days (impulse response). The fiscal cost of the Quota Tariff, the forgone tariff revenue, is substantial for the vegetable designations as well; since these designations have no identified effect on prices, refraining from them would save a substantial fiscal cost, whereas for the fruits this cost came with an identified price reduction.

Keywords: Quota Tariff, Price Stabilization, Local Projection Difference-in-Differences, Agricultural Trade Policy

JEL Classification: F13, F14, Q17, C23

1 Introduction

This paper studies Korea's Quota Tariff system, grounded in *Article 71 of the Customs Act*, focusing on the tariff reductions aimed at stabilizing agricultural prices. The Quota Tariff is a flexible tariff instrument: by Presidential Decree, the government can adjust both the reduced tariff rate and the volume of imports eligible for that rate. This allows the government to charge a rate below the Basic Tariff Rate, supporting the smooth supply of goods and promoting domestic price stability. Moreover, a quantity limit is not an essential element of the system, which allows a temporary tariff reduction to be granted without any volume restriction. This sets the Quota Tariff apart from the Tariff-rate Quota (TRQ) operated under the World Trade Organization (WTO) and free trade agreements (FTAs). The TRQ guarantees the market access a country has committed to under the WTO Agreement on Agriculture or an FTA, applying a low tariff rate up to a specified volume and a higher rate to imports beyond it, and so pursues market opening and the protection of domestic industry at the same time. Korea's Quota Tariff, by contrast, adjusts tariffs flexibly for domestic policy purposes independently of such international obligations, and is in this respect a far more flexible instrument than the TRQ. To keep the two distinct, this paper uses the term 'Quota Tariff' to refer to Korea's system.

Prior studies have focused mainly on the trade and protection effects of the TRQ (Loginova et al., 2021; Schaefer and Wolf, 2025), whereas empirical analyses of the Quota Tariff are scarce; this paper fills that gap by analyzing the causal effect of the Quota Tariff on consumer prices. The gap extends beyond the choice of subject to the methodology. The tariff measures imposed by the TRQ family and by the Quota Tariff are temporary, and they therefore generate a non-absorbing treatment: one that starts and ends repeatedly within the sample period. Yet, to the best of our knowledge, only one study, Loginova et al. (2021), estimates the causal price effects of such institutions directly on non-absorbing data. This study instead estimates the effects dynamically from the non-absorbing treatment path itself, using a daily treatment record that reflects even the lapses caused by quota exhaustion; section 2 reviews this methodological gap.

Finally, the study also has important implications for Korea's Quota Tariff policy on agricultural products, where debate over the price-stabilizing effect of the Quota Tariff has persisted for years. As of 2024, coverage of agricultural products had

expanded considerably: the number of covered products rose from 20 in 2020 to 72 in 2024, and total tariff reductions reached about one billion dollars, a substantial fiscal cost (Jang, 2025). Since about 66 percent (%) of these measures were deployed to stabilize retail prices, it is important to assess whether the policy actually achieves its original goal.

To answer this question, this paper constructs a daily panel of retail prices covering 41 agricultural products from January 2021 through March 2025: eleven products that had the Quota Tariff in force at least once during the sample period (six leafy and root vegetables and five fruits) and 30 products that never did. Designations begin on different dates across products (staggered adoption), a lapsed designation can be renewed on the same product (non-absorbing treatment), and the depth of the tariff reduction differs widely (heterogeneous treatment intensity). Conventional difference-in-differences (DiD) estimators are either biased under this structure or presuppose absorbing treatment. This paper therefore adopts local projection difference-in-differences (LP-DiD), which accommodates non-absorbing treatment (Dube et al., 2025). For each of the two treatment definitions, treatment status (binary) and treatment intensity (continuous), the estimation targets the effect of treatment while the tariff reduction remains in force (persistent treatment) and the total effect of initiating a Quota Tariff designation (impulse response). The analysis assumes the stable unit treatment value assumption (SUTVA): one product's outcome is unaffected by another product's treatment. Section 5.2 relaxes this assumption for the products most likely to be affected, candidate substitutes and complements of the ever-treated products, by removing them from the comparison pool, and the main estimates keep their signs and 5% significance.

For the vegetable group, composed of leafy and root vegetables, the estimated retail price reductions are insignificant in every specification, whereas for fruits a significant reduction is found. The estimated response of fruit retail prices is about -0.78% per one percentage point (%p) of tariff reduction at 150 days after onset (the effect while the tariff reduction remains in force) and -0.49% to -0.63% at 250–273 days after onset (the total effect of initiation). Full and immediate pass-through would imply 0.77% to 0.94% per 1%p; the fruit estimates therefore indicate that a substantial share of the tariff reduction reached consumers, at the shorter horizon almost all of it.

The Quota Tariff lowers the tariff only on imports, so its effect can reach retail

prices only through the imports whose tariff is reduced. In 2021, imports on the tariff lines later covered by designations made up all of the supply of the four tropical fruits in the sample (mango, banana, avocado and pineapple), for which no domestic production is recorded, but only a small share of the supply of every ever-treated vegetable except carrot. The ever-treated fruits are also far more import-dependent than most of the ever-treated vegetables. Both contrasts are consistent with the difference between the fruit and vegetable results, but both measures differ between the two product groups, so the estimates cannot test whether either is the cause of that difference; this paper presents the link between these contrasts and the results as an interpretation (section 5.3).

The fiscal cost of a Quota Tariff, the forgone tariff revenue, is substantial for the vegetable designations as well, because carrot and onion are imported in large volumes, yet these designations have no identified effect on prices, whereas for the fruits this cost came with an identified price reduction. These results suggest that refraining from the vegetable designations, for which no price-reducing effect is identified, would save a substantial fiscal cost, and, if this interpretation holds, that the Quota Tariff is effective as a price-stabilization instrument for products sold at retail only as imported produce, such as these tropical fruits.

The data and code needed to replicate this study are publicly available.¹ Section 2 presents the institutional background and related literature, section 3 the data, section 4 the LP-DiD estimates, section 5 the robustness of the estimates and the interpretation of the difference between the fruit and vegetable results, and section 6 the conclusions and policy implications.

2 Institutional Background and Related Literature

The TRQ originates in the Uruguay Round: the WTO Agreement on Agriculture converted non-tariff barriers into tariffs and, in exchange, introduced the TRQ to guarantee minimum market access. It has since become a principal instrument for opening agricultural markets not only under that agreement but also under FTAs. Tariffs, however, also serve domestic policy objectives. Institutional designs differ

¹ https://github.com/jayjeo/github_TRQ

across countries, but temporary tariff adjustments to relieve supply shortages or stabilize prices can be found in a number of them. Korea's Quota Tariff, the subject of this study, is one such instrument, operated under *Article 71 of the Customs Act* separately from the TRQs of the WTO and FTA frameworks. Because it operates by Presidential Decree rather than through the legislative process of the National Assembly, the tariff rate and the volume of imports eligible for that rate can be adjusted quickly and flexibly as market conditions change.

The existing literature has mainly analyzed the economic effects of TRQs administered within the WTO and FTA frameworks. Loginova et al. (2021) analyze how Switzerland's administration of agricultural TRQs affects domestic price formation. Swiss protection periods cover each vegetable's main harvest season, during which prohibitive over-quota tariffs largely block imports. The authors find that these protection periods raised Swiss producer prices of most vegetables by more than 20% relative to those in neighboring countries, and by more than 50% for some. These price effects, however, were not uniform across vegetables.

In addition, Schaefer and Wolf (2025) analyze the reform of Canada's dairy TRQ system. After a dispute-settlement panel under the United States–Mexico–Canada Agreement ruled against Canada's practice of reserving dairy TRQ volumes for processors, the revised allocation generated additional trade expansion for some dairy products. The literature has thus focused mainly on the market-access and trade effects of TRQs. By contrast, empirical evidence on the consumer-price effects of tariffs used as an instrument of domestic price stabilization remains insufficient. For Korea's Quota Tariff, Song (2023) analyzes the relationship between the import price excluding tariffs and consumer prices, but this approach falls short of directly identifying the policy effect of the Quota Tariff itself.

The tariff measures imposed by TRQ-type institutions and by the Quota Tariff are temporary. Protection phases of seasonal TRQs open and close with the domestic harvest calendar (Loginova et al., 2021; Hillen, 2019), safeguard tariffs are triggered when cumulative imports cross a threshold and lifted afterwards (Livingston et al., 2025), and a Quota Tariff designation lapses when its notified period ends or its quota is exhausted. The resulting treatment is non-absorbing: it can switch off and later switch back on (Bach et al., 2025; Dube et al., 2025). To the best of our knowledge, only

Loginova et al. (2021) estimate the price effects of such measures directly on non-absorbing data, pooling the protected and unprotected phases; other studies either treat the policy as an absorbing treatment (Flaaen et al., 2020; Fajgelbaum and Khandelwal, 2026) or examine outcomes other than prices (Hillen, 2019; Livingston et al., 2025). This paper addresses the gap by applying the existing LP-DiD framework for non-absorbing treatment (Dube et al., 2025), adapted to a continuous treatment intensity.

A tariff change reaches retail prices in stages. The 2018 United States (U.S.) tariff increases raised the tariff-inclusive prices of the targeted imports by the full amount of the tariffs (Amiti et al., 2019; Fajgelbaum et al., 2020), whereas retail prices in U.S. stores responded less, which suggests that retail margins fell (Cavallo et al., 2021). How much of a tariff change reaches the retail price of an imported good thus depends on pass-through at the border and on the adjustment of distribution margins. For the ever-treated fruits, the estimates of this paper indicate that a substantial share of the tariff reduction reached consumers: the retail response 150 days after onset while the reduction is in force, about 0.78% per 1%p of tariff reduction, is close to the 0.77% to 0.94% implied by full and immediate pass-through (section 4.5), and wholesale prices fell as well, with no significant change in the retail-to-wholesale margin (section 5.3).

A second channel works through domestically produced goods that compete with imports: the imports-as-market-discipline hypothesis holds that import competition disciplines the pricing of domestic producers (Levinsohn, 1993), and a tariff can raise the prices of competing domestic brands and of complementary goods that it does not cover (Flaaen et al., 2020). The retail prices of the four ever-treated tropical fruits in Korea are prices of the imports whose tariff a designation lowers, whereas those of the ever-treated vegetables are prices of both domestic and imported produce, and a designation can reach the domestic produce only through this second channel. This paper detects no price-reducing effect for the vegetables. In 2021, imports on the tariff lines later covered by designations made up all of the supply of the four tropical fruits, for which no domestic production is recorded, but only a small share of the supply of every ever-treated vegetable except carrot; this contrast in the share is consistent with the difference between the fruit and vegetable results. However, the share, like import dependence, differs between the two product groups, so the estimates cannot test whether it is the cause of that difference (section 5.3).

Fresh produce is perishable and its domestic supply is seasonal: Swiss seasonal TRQs protect domestic producers from competing imports during the harvest season (Hillen, 2019). Yield shocks also shift the supply and prices of agricultural commodities (Roberts and Schlenker, 2013). For the vegetables grown in Korea, domestic weather therefore drives both prices and the timing of designations. Consistent with this, vegetable prices in this paper rise before designation, and the rise is larger without the climate covariates (section 4.4), whereas the predominantly imported ever-treated fruits, for which Korean weather is not the relevant supply shifter, show no systematic rise. This is why only the vegetable specification includes climate covariates (section 4.3).

3 Data Construction

The dataset constructed for this study is a product-level daily panel covering 41 agricultural products from January 1, 2021, through March 31, 2025. Daily retail prices, obtained from *Nongnet*, serve as the dependent variable of the causal analysis. The data on Quota Tariff status, designation start and end dates, and treatment intensity by product (the most critical explanatory variables in the regressions) were compiled from the public records published under the *Regulations on the Application of Quota Tariffs under Article 71 of the Customs Act*. This section describes how these variables, and the covariates for two important confounders, are constructed; the remaining covariates are described briefly where they enter the analysis in section 4.

Section S1 of the Online Supplementary Material presents the summary statistics underlying the LP-DiD analysis: table S1 covers every variable entering the regressions, while tables S2 and S3 report the annual import volumes and values of the 41 products. Because these absolute quantities do not show how large imports are relative to domestic supply, table S4 compares the pre-treatment import dependence of ever-treated and never-treated products, measured as the share of imports in domestic availability over 2019–2021.

3.1 Selection of Treated and Controls

Since the analysis relies on LP-DiD, selecting the treated and control groups is essential. A structural feature of LP-DiD, however, is that a treated item can also serve as a control

at some horizons (h), as section 4.2 discusses in detail. The selection is therefore stated more precisely in terms of an ever-treated and a never-treated group. An ever-treated item is one that had the Quota Tariff in force at least once during the sample period, and depending on the horizon it functions sometimes as treated and sometimes as a control; a never-treated item never had the Quota Tariff in force and functions only as a control.

The inclusion and exclusion rules deserve a full account, because it matters for the credibility of the design that group membership is determined by observable criteria, not by arbitrary selection on our part. The analysis is confined to agricultural products: livestock, fishery, and forest products lie outside its scope, which excludes foods such as chicken and eggs, as well as mushrooms, which official statistics group with agricultural products but which are legally classified as forest products.² Items whose daily retail price series (the dependent variable) are unavailable or mostly missing are likewise excluded: broccoli, cherry, Korean melon, mustard greens, orange, peach, strawberry, and sweet persimmon. Potato, grape and rice are also excluded (section S1): the Quota Tariff on potato covers only potatoes for chip manufacturing, whereas retail prices are observed only for table potatoes; grape has no retail price series over its event window; and rice is managed under the government’s priority food policy.

Among the 41 products so selected, any item for which the Quota Tariff was in force at least once during the sample period (January 1, 2021, through March 31, 2025) is classified as ever-treated and the rest as never-treated. The eleven ever-treated items are listed by group. The leafy and root vegetables are napa cabbage, cabbage, radish, carrot, onion, and green onion; the fruits are mango, kiwifruit, banana, avocado, and pineapple. The remaining 30 products are all never-treated; panel C of table S1 lists them, divided into never-treated vegetables and other crops, and never-treated fruits.

3.2 Quota Tariff Treatment Status for Ever-Treated Items

Table 1 reports, on a daily basis, when each ever-treated item was under the Quota Tariff. Border lines indicate monthly boundaries; white cells denote days on which the Quota Tariff was not in force (*not-treated status*), whereas black cells denote days on which it was in force (*treated status*). Data prior to 2022 are omitted for space, although

² *Forestry and Mountain Villages Development Promotion Act.*

daily treatment status was constructed for all products from 2015 onward; designations were exceedingly rare from 2015 through 2022 and increased markedly from 2024 onward. The treatment path used in the estimation is the status recorded in this table with a single adjustment: year-end gaps of at most 30 days within a notice sequence (nine gaps) are bridged.³ Gaps longer than 30 days are never bridged and are used exactly as recorded.

Table 1: Quota Tariff Treatment Status

	2022												2023												2024												2025											
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8				
Napa Cabbage																																																
Cabbage																																																
Radish																																																
Carrot																																																
Onion																																																
Green Onion																																																
Mango																																																
Kiwifruit																																																
Banana																																																
Avocado																																																
Pineapple																																																

The remainder of this subsection details the criteria by which the black and white cells in table 1 were determined. This matters because the treatment status is the foundation of the LP-DiD analysis: it is what identifies the clean controls and clean treated units at the core of the design, and it must therefore be constructed accurately from the factual record. As the table shows, the treatment status in this study is ‘non-absorbing.’ A configuration in which white cells appear first chronologically and, once transitioning to black, never revert to white is termed ‘absorbing’ (Bach et al., 2025); configurations permitting reversion from black to white are termed ‘non-absorbing’ (Dube et al., 2025), which allows for indefinite white-black-white-black repetition.

³ The bridge is dictated by the design: the gaps run 13 to 23 days, far shorter than the 120-day washout, so treating each renewal as a new onset would make every such re-onset dirty and split a single policy episode into a short event and an unusable orphan segment (section 4.2.2).

A first distinction is between designations announced with a quota limit and those announced without one. Under a limited designation the tariff relief is available only until cumulative imports, counted from the start of the designation, reach the quota; once the quota is exhausted, the relief lapses even though the announced period has not ended, and the item must be regarded as reverting to not-treated status. Two notions of duration must therefore be kept apart: the *designated period*, announced in advance in the government notice, and the days on which the Quota Tariff is actually *in force*, which is what the treated status in table 1 records. Under an unlimited designation the two coincide and the relief runs uninterrupted to the end of the designated period.

Determining the exhaustion date requires care, because the cumulative imports counted against the quota are not the item’s total imports but only the imports cleared through customs under the Quota Tariff. The Korea Customs Service discloses this narrower series by item, but at monthly rather than daily frequency, which reveals the month in which the quota was exceeded but not the exact day; apart from this limitation it is the most authoritative source available (Method A). Alternatively, the exact crossing day can be computed from the item’s monthly total import volume under the assumption that each month’s imports are distributed uniformly across its days (Method B). The drawback is that Method B rests on total imports rather than only those cleared under the Quota Tariff, and it therefore overstates the relevant volume. This study combines the two methods to arrive at the most defensible exhaustion date for each episode.

Pineapple illustrates the procedure. Of the six Quota Tariff designations announced for it during the sample period, only one ended in exhaustion: under the designation running from August 25 to December 31, 2023, the quota was crossed on October 9, 2023, making October 8 the last day on which the Quota Tariff was in force, and the cells from October 9 onward within the designated period are white in table 1 (not-treated status). In the other five cases the quota never bound within the sample period. Table S5 in section S2 sets out the announced period and the quota of each of the six designations, and that section works through every case.

3.3 Anticipation of the Quota Tariff Policy

The extent to which agents could anticipate a designation also matters for identification.

Full anticipation would lead importers and distributors to defer imports until the lower tariff took effect and to run down inventories ahead of the expected price decline, and such behavior would affect the estimates. For the agricultural Quota Tariff, however, the scope for anticipation is limited. Unlike designations for industrial raw materials and similar items, for which the government announces an annual Quota Tariff operation plan in advance, agricultural designations are mostly emergency responses to current market conditions. Some measures took effect with no advance legislative notice at all; where notice was given, a lead of about one month occurred in only a few cases, and for most measures the interval was shorter still. Once the decree was promulgated, enforcement typically began immediately or within one to five days. The policy thus leaves economic agents little practical room to adjust before it takes effect. The analysis accordingly does not treat the legislative notice as a separate policy shock, and identifies the policy effect from the date each designation takes effect.

3.4 Treatment Intensity

The LP-DiD analysis is conducted for two treatment definitions. Treatment status targets the effect on consumer prices of a Quota Tariff designation, captured in the regressions by a dummy variable for its onset; treatment intensity targets the effect of a designation on consumer prices per 1%p of the tariff reduction it generates, captured by a continuous variable measuring how far the designation lowers the tariff rate relative to its pre-designation level. Computing the latter requires two pieces of information: the tariff rate charged while the Quota Tariff is in force, which is read directly from the government notice, and the tariff rate that would prevail in its absence, a rate we denote the ‘applied tariff rate.’

Deriving the applied tariff rate requires complex calculations across tariff agreements and Basic Tariff Rates for each country, product, and month, a task demanding specialized expertise in customs administration. Multiple rates coexist for each Harmonized Tariff Schedule of Korea (HSK)⁴ code, including Basic Tariff Rates and Preferential Tariffs under an FTA, and *Article 50 of the Customs Act* governs the

⁴ The HSK extends the internationally harmonized six-digit subheadings of the Harmonized System to ten-digit national tariff lines, at which the Korean government defines individual products and assigns tariff rates. The four digits beyond the sixth are specific to Korea and do not correspond to the tariff lines of other countries.

priority order in which they apply. Even for identical products, the rate actually applied may vary with the country of origin, product specifications, and importer circumstances. No single statutory rate can therefore represent the tariff burden on a product, and an indicator such as the applied tariff rate is needed.

This indicator is constructed from actual import records. Country-specific monthly import volumes by product⁵ drawn from the *Korea Agricultural Trade Information* (KATI) serve as the weights, since the rate that applies to the same product differs with the exporting country. For each country and month, the ‘most favorable tariff rate’ applicable to that country’s imports was determined by following the ‘Tariff Rate Priority Structure’ established under the *Customs Act* and presented in section S3. The monthly applied tariff rate is the average of these country-specific rates weighted by import volume; repeating the calculation for every month from January 2021 through March 2025 yields a product-level panel of monthly applied tariff rates. This procedure cannot verify directly whether Preferential Tariffs were applied in individual transactions, but by combining the structure of the tariff system with observed import patterns it provides a meaningful estimate that closely approximates the tariff burden actually incurred; section S3 notes one further limitation, concerning Specific Duties.

Two worked examples show what this produces. For banana the volume-weighted applied tariff rate was approximately 23.3% in July 2024, and fell to approximately 14% to 15% in the first quarter of 2025 after the Korea-Philippines FTA entered into force. For fresh unpeeled onion the rate swings between 50% and 135% from month to month: Chinese fresh unpeeled onion, the bulk of imports, derives no practical benefit from the FTA Tariff Concessions, leaving the WTO Tariff Concession to apply, at the Recommended Tariff Rate within the Market Access Quota and at the non-recommended rate beyond it. Section S4 works through two examples, banana and onion, country by country and plots the resulting monthly series of applied tariff rates in figure S1.

3.5 Removing Seasonality

This paper uses deseasonalized prices as the dependent variable, a choice justified as

⁵ Ten-digit HSK code.

follows. Seasonality is removed through a regression-based adjustment using calendar dummy variables and Fourier terms (Pierce et al., 1984), designed to preserve the treatment effects. For daily series, standard official seasonal-adjustment tools (e.g., X-12-ARIMA/X-13ARIMA-SEATS) are not directly applicable because they are designed for monthly or quarterly data and assume fixed 12-month or 4-quarter cycles (Mollins and Lumb, 2024). The natural alternative, seasonal-trend decomposition based on loess (STL), is a non-parametric method that has been applied extensively to weekly and daily data (Cleveland et al., 1990); because it relies on non-parametric smoothing, however, part of a trend or policy shock may be absorbed during the decomposition, and treatment effects removed at the seasonal-adjustment stage would distort the subsequent DiD analysis. The regression-based adjustment with calendar dummies and Fourier terms is chosen precisely because it removes the seasonal repetition while leaving the policy effect intact.

One might nevertheless object that, because the Quota Tariff is deployed precisely to suppress prices when they surge seasonally, the analysis should rely on the original prices. This objection can be countered on two grounds. (a) The pure effect of the Quota Tariff survives deseasonalization: the timing of Quota Tariff designations in Korea is far from a regular seasonal pattern, as table 1 shows, and the adjustment removes only recurring patterns, not genuine shocks, so the effect becomes even more transparent once the seasonal repetition is stripped away. (b) Using the original prices within the LP-DiD framework would obscure that effect, because each item has its own seasonal frequency and amplitude, making treated and control items not directly comparable.

The detailed implementation of the deseasonalization is documented in section S5, where figure S2 illustrates the procedure with the original and seasonally adjusted series of spinach retail prices.

3.6 Confounding Effects

The Quota Tariff is not the only instrument the government deploys against agricultural price surges. Two further instruments share both its trigger (supply shortfalls and surging prices) and its purpose of stabilizing consumer prices: temporary expansions of WTO Market Access Quotas and releases of government-stockpiled produce. Either can move retail prices at the same time as a Quota Tariff designation, and leaving them

uncontrolled risks conflating their effects with the effect under study. The estimation therefore includes covariates for both, collected in the blocks Q_{it} and R_{it} of the covariate vector in section 4.3.

The first instrument operates within the conventional WTO-style TRQ, from which section 2 distinguishes the Quota Tariff: imports within a scheduled Market Access Quota enter at a low in-quota rate, while imports beyond it face a substantially higher over-quota rate. Only the items for which Korea’s WTO concessions schedule a quota (onion, garlic, and ginger among them, but not banana, avocado, or napa cabbage) can experience an expansion, which the government may grant temporarily when supply–demand imbalances intensify, as it has done for sesame and onion. Since an expansion admits a larger import volume at the low in-quota rate, it is a tariff-relief measure deployed on the same grounds as the Quota Tariff. The expansion history is compiled from the enactment records (section S1 gives the coding rule).

The block Q_{it} accordingly contains three covariates: a coverage dummy for items under a WTO TRQ; the expansion ratio in force on day t (the additional volume granted by the active expansion relative to the scheduled quota, equal to zero when no expansion is in force); and the expansion ratio interacted with the tariff gap between the over- and in-quota rates, which scales each expansion by the size of the tariff gap it relieves. The coverage dummy separates items with no quota to enlarge from covered items for which the government chose not to expand the quota.

The second instrument is the release of government stockpiles: the government stores staple produce in normal times and releases it when supply tightens or prices surge, with napa cabbage, radish, and dried red pepper among the prominent cases. The block R_{it} contains two covariates built from the monthly release records (section S1): the inverse hyperbolic sine of the average release over the trailing window $[t - 130, t - 31]$, which compresses the heavy right tail of the volume distribution while keeping the many zeros (Bellemare and Wichman, 2020), and an incidence dummy equal to one when any release falls in that window. The window closes before the month containing day t , so that monthly release information from after the Quota Tariff treatment date cannot enter.

In the daily panel of the eleven ever-treated products, the quota-expansion channel never coincides with the Quota Tariff: onion is the only one covered by a WTO TRQ,

and its single expansion episode (July–December 2023) does not intersect its designation (August 2022–February 2023). The expansions of the period fall instead on never-treated products in the comparison group, whose movements the Q_{it} covariates absorb.

The stockpile channel, by contrast, overlaps the treatment substantially: the three stockpiled ever-treated products (onion, napa cabbage, and radish) all began their longest designation episodes in months with active releases, and releases were under way in the same month on 70% (573 of 815) of their days under the Quota Tariff. Within-product correlations between daily Quota Tariff status and the two release covariates are positive though moderate, 0.11 to 0.16 over the eleven products and 0.23 to 0.33 within the stockpiled three. Omitting the release covariates would therefore risk crediting the Quota Tariff with the stabilizing effect of the releases.

4 Causal Analysis

This section estimates the causal effect of the Quota Tariff on retail prices. Section 4.1 explains why the treatment structure calls for LP-DiD, and sections 4.2 and 4.3 set out the estimands, the sample restrictions and the estimating equations. Section 4.4 probes the conditional parallel-trends assumption, and sections 4.5 and 4.6 report the estimates, as dynamic response graphs and at selected horizons.

4.1 Why Local Projection Difference-in-Differences

Three features of the Quota Tariff, established in section 3, jointly determine the choice of estimator. Adoption is staggered: products enter treatment at different dates. Treatment is non-absorbing in the sense set out in section 3.2: a designation lapses at its notified end date or, for quota-limited notices, once the quota is exhausted, and the same product may be designated again later. Finally, treatment intensity is heterogeneous: the implied tariff reduction ranges from 1.5%p to 69.9%p across products (table 2). The analysis accordingly asks two questions on the same panel: whether Quota Tariff adoption as such lowers retail prices (a binary treatment-status question), and by how much retail prices fall per 1%p of tariff reduction (a continuous treatment-intensity

question).

Conventional two-way fixed effects (TWFE) event-study regressions are ill-suited to this structure. When treatment turns on and off and effects are heterogeneous, they implicitly use previously treated observations as comparisons even though the lagged effects of an earlier episode may still be unfolding. The resulting contamination can severely bias the estimates, including through negative weighting (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfœuille, 2020). Retail prices of pineapple in early 2024 are a poor counterfactual, because the effects of its 2023 episode may persist well after quota exhaustion ended it.

The recent heterogeneity-robust estimators confront this contamination by restricting the controls to observations that are not yet treated or never treated (Roth et al., 2023; de Chaisemartin and D’Haultfœuille, 2023). But the interaction-weighted estimator of Sun and Abraham (2021) is constructed for the absorbing case and characterizes each unit by a single initial-treatment date, as does the irreversibility-of-treatment assumption of Callaway and Sant’Anna (2021). In contrast, the Quota Tariff is non-absorbing in *both* the treatment-status and the treatment-intensity specifications. Recoding treatment as an ever-treated indicator would change the estimand to the effect of *having ever* been designated and misclassify terminated and quota-exhausted periods as treated; relabeling each onset as a fresh absorbing cohort fails too, because several episodes adjoin so closely that any inter-episode “pre-period” would be contaminated by the preceding one.⁶

This study therefore employs LP-DiD (Dube et al., 2025), which is constructed for non-absorbing treatments: each horizon-specific regression compares newly treated observations only with contemporaneously ⁷ clean comparisons. Three further properties make it the natural choice here. It conditions directly on the pre-determined time-varying covariates, which the specification below requires. It nests the binary and the intensity treatments in one framework, so that differences between the two sets of estimates reflect the treatment definition rather than a change of estimator family. And it extends naturally to a treatment that is, as here, simultaneously non-absorbing and

⁶ The improved estimators for staggered designs proposed by Sun and Abraham (2021), Callaway and Sant’Anna (2021), Borusyak et al. (2024), and Gardner (2022) all presume that treatment, once switched on, stays on.

⁷ At the same horizon.

continuous: Dube et al. (2025) flag continuous treatment as a natural extension of their binary non-absorbing treatment, and this study implements that extension through a pre-determined intensity scalar (see Dube and Lindner, 2024 for an LP-DiD application with non-absorbing treatment). The credibility of LP-DiD rests not on the regression equation but on the discipline it imposes: strict rules about which observations may serve as comparisons.

4.2 Core Design Elements of Local Projection Difference-in-Differences

The LP-DiD design has two components: a choice of estimand, and a pair of sample-selection rules—clean controls and clean treated units—governing which observations enter the horizon- h regression.

4.2.1 Two Estimands: Persistent Treatment versus Treatment Initiation

The *persistent-treatment* scheme asks what happens to prices while a designation is in force: it keeps a treatment event in the horizon- h regression only while the treatment persists. The *treatment initiation* scheme asks what the total effect of having initiated a designation is, including whatever happens after the episode ends or the product re-enters treatment; it follows every initiation across all horizons, leaving the post-initiation path as it actually unfolded. The coefficient path is thus an impulse response function (IRF): the average effect of treatment *initiation* h days later. This is the impulse-response-type estimand discussed by Dube et al. (2025), corresponding to the actual-versus-status-quo event-study effects formalized by de Chaisemartin and D’Haultfœuille (2026). Neither estimand is more correct than the other; Dube et al. (2025) regard the choice as a matter of the research question, and here the two are complements. The persistent-treatment estimates address whether the policy lowers prices while it operates. The impulse-response estimates bear on whether initiation effects outlive the designations themselves, a question of first-order policy interest for a temporary instrument, but do not by themselves establish that the effects do so. The two schemes differ in exactly one sample condition on the treated side.

4.2.2 Clean Controls and Clean Treated

For a regression at horizon h , the comparison group at date t consists of product–day observations that have never been treated up to t and remain untreated through $t + h$. This “clean control” design, named in the stacked event-study literature on minimum wages (Cengiz et al., 2019) and formalized as a flexible clean control condition by Dube et al. (2025), follows from the contamination problem above. Units whose own treatment dynamics are still unfolding cannot reveal the untreated counterfactual. The same comparison group serves both schemes, *persistent-treatment* and *treatment initiation*, and the implementation here is deliberately stricter than the theory requires. The window-based condition of Dube et al. (2025) would readmit a previously treated product once its dynamics have stabilized; this study instead removes a product from the control pool permanently after its first treatment, which leaves an ever-treated product contributing clean control observations only before its first onset. Since 30 of the 41 products are never treated, clean controls remain abundant at every horizon.

A parallel discipline applies to the treated side, where the two schemes, *persistent-treatment* and *treatment initiation*, diverge. The *washout* condition requires the product to have been untreated throughout the 120 days preceding the onset, that is, over $[t - 120, t - 1]$. This is what makes an onset *clean*: because no treatment precedes it within the window, the dynamics attributed to the event reflect that event alone rather than the residue of a recent previous episode. The condition implements the assumption of Dube et al. (2025) that the effects of a past episode stabilize within L periods, where $L = 120$ in this study. Since cleanliness concerns the past, the washout applies to every specification, the impulse-response ones included. In contrast, the *stay-on* condition is not about cleanliness. It retains a treated event at horizon h only if the product is treated on every day of $[t, t + h]$, which is what defines the persistent-treatment estimand, the effect of treatment still in force at $t + h$. It therefore applies to the persistent-treatment specifications only, and the impulse-response ones drop exactly this condition and nothing else. Together the two conditions reproduce the treated-side restriction that Dube et al. (2025) impose for non-absorbing treatments, and the same design logic runs through the stacked event-study literature (Cengiz et al., 2019; Wing et al., 2024; Dube and Lindner, 2024).

The treated and control sides are thus disciplined asymmetrically: a treated event

needs a clean *recent history*, whereas a control observation needs a clean *entire history*. Pineapple shows that the washout is no formality: its onset of January 19, 2024, fails the condition, falling only 102 days after the preceding episode ended with quota exhaustion on October 8, 2023 (section 3.2). Nor can any pineapple observation after its first designation serve as a control, since the product’s treatment history is no longer clean. Table S6 in section S6 classifies every onset—three fail the washout in all—and each classification follows mechanically without discretion. That section also discusses the one clean onset that falls within the 273-day window over which the impulse-response specification follows an earlier onset, and why the 120-day washout is nonetheless retained. The treated-side conditions are used exactly as prescribed by Dube et al. (2025); only the control-side condition is stricter.

4.2.3 Horizon Coverage and Limits of the Two Estimands

The stay-on condition has a mechanical consequence that shapes how far the persistent-treatment estimates can be read. An event whose treatment spell lasts s days satisfies the stay-on condition only for horizons $h < s$, which removes shorter-spell events one by one as h grows. Among the eleven ever-treated products, nine vegetable and eight fruit events survive the washout screen at $h = 0$. Both sets then shrink step by step as the spells of individual events end (table S6 in section S6). As h grows, the estimates therefore compare a changing collection of events, a “composition effect” inherent to horizon-specific estimation (Dube et al., 2025). At exactly $h = 156$, the day the cabbage event drops out, the vegetable coefficient in the treatment-status specification jumps from +0.4 to +18.2 log points and immediately turns statistically significant, while the surviving treated set comes to be dominated by the radish and napa cabbage events of 2024, whose post-treatment windows coincide with the severe 2024 supply disruptions in those markets. A discontinuity that arrives on the day of a compositional break, rather than growing out of the preceding dynamics, is a composition artifact, not a treatment dynamic. The persistent-treatment graphs are therefore truncated at $h = 150$, a conservative margin below the break at 156, with the underlying estimates for all horizons up to 273 retained in the replication files. Attrition, however, is not a defect of the design but a direct consequence of the stay-on condition, which is precisely what defines the persistent-treatment estimand. The impulse-response specifications, having

no stay-on condition, are free of attrition: their treated counts stay at nine and eight across all horizons, and the impulse-response graphs extend to $h = 273$, the maximum the sample supports. The 2022 pineapple treatment event, for instance, enters the persistent-treatment regressions only through $h = 51$, where its tariff cut ends, but the impulse-response regressions through $h = 273$.

The impulse-response estimand, although free of attrition, has a limitation that should be stated plainly. Treatment durations differ widely across events (spell lengths range from 45 to 585 days, the longest right-censored by the end of the sample), and the impulse-response coefficient averages over initiations without adjusting for the number of treated days each event accumulates within the window. It is therefore a per-initiation effect, not a per-treated-day effect. The same convention governs the intertemporal framework of de Chaisemartin and D’Haultfœuille (2026), whose event-study effects are likewise defined by the change in treatment alone, with no duration term entering the estimation. Read this way, the impulse-response coefficient answers the question a policymaker faces at the moment of designation: what initiating a Quota Tariff sets in motion for retail prices h days ahead.

4.3 Regression Specification

Following the pre-mean-differenced (PMD) LP-DiD of Dube et al. (2025), the dependent variable at horizon h is the log retail price h days after the event, measured against the average log price over the 120 days preceding the event:

$$\bar{y}_{i,t-1} \equiv \frac{1}{120} \sum_{s=1}^{120} \ln P_{i,t-s}, \quad \tilde{\Delta}^h \ln P_{it} \equiv \ln P_{i,t+h} - \bar{y}_{i,t-1}, \quad -500 \leq h \leq 273, \quad (1)$$

where i and t denote product and calendar day, and P_{it} is the seasonally adjusted retail price (KRW/kg) of section 3.5. The pre-mean baseline is fixed before the event and is not an ad hoc choice: Dube et al. (2025) develop this PMD LP-DiD on the efficiency argument of Borusyak et al. (2024) that the pre-event level is estimated more precisely from many pre-treatment days than from the single day $t - 1$. The `lpdid` package in *Stata* and *R* implements it as a built-in option (Busch and Girardi, 2024). The argument

has particular force in daily retail prices, where a single-day baseline would transmit that day's idiosyncratic movement into the outcome at every horizon.⁸ Since equation (1) differences each observation against its own pre-event level, product fixed effects are differenced out by construction.

For each horizon h , the treatment-status (binary) specification is the cross-sectional regression:⁹

$$\tilde{\Delta}^h \ln P_{it} = \beta^h (D_i \cdot \Delta D_{it}) + T_t + X_{it} + \varepsilon_{it}^h, \quad (2)$$

estimated, for $h \geq 0$, on the sample

$$\begin{aligned} \mathcal{S}_{\text{base}}^h = & \underbrace{\{(i, t): \Delta D_{it} = 1, D_{i,t-s} = 0 \forall s \in [1, 120], D_{i,t+j} = 1 \forall j \in [0, h]\}}_{\text{clean treated: fresh entry, remaining treated through } t+h} \\ & \cup \underbrace{\{(i, t): D_{is} = 0 \forall s \leq t, D_{i,t+j} = 0 \forall j \in [1, h]\}}_{\text{clean control: no history, untreated through } t+h}. \end{aligned} \quad (3)$$

Here $D_{it} \in \{0, 1\}$ is the daily Quota Tariff treatment status of table 1 and D_i indicates ever-treated products. T_t are calendar-day fixed effects, so that every event-time comparison is simultaneously a same-calendar-date comparison, absorbing market-wide movements. The two sets in equation (3) are built separately rather than by matching treated to control observations one-to-one, and their dates t need not coincide: the horizon- h regression pools both into one sample, and the same-calendar-date comparisons are carried out not by the sample definition but by the calendar-day fixed effects. For $h < 0$ the dependent variable is the log retail price $|h|$ days before the event, measured against the same pre-event mean; at these pre-onset horizons the forward conditions in equation (3) are dropped, namely the stay-on condition $D_{i,t+j} = 1 \forall j \in$

⁸ The 120-day window matches the washout length and was fixed ex ante, in line with the recommendation of Dube et al. (2025) that the reference period be chosen according to the characteristics of the application, independently of the estimation results. The size of the efficiency gain depends on the serial correlation of untreated outcomes (Borusyak et al., 2024; Dube et al., 2025). PMD does not by itself guarantee parallel trends; the pre-event coefficients over $[-120, -1]$ reported in section 4.5 remain the relevant diagnostic.

⁹ Although the data used in this study form a panel, LP-DiD selects every observation satisfying the clean-control and clean-treated conditions and then estimates the equation on that sample as a cross-sectional rather than a panel regression.

$[0, h]$ on the treated side¹⁰ and the future no-treatment condition $D_{i,t+j} = 0 \forall j \in [1, h]$ on the control side:

$$\mathcal{S}_{\text{base}}^h = \underbrace{\{(i, t): \Delta D_{it} = 1, D_{i,t-s} = 0 \forall s \in [1, 120]\}}_{\text{clean treated: fresh entry, washout only}} \cup \underbrace{\{(i, t): D_{is} = 0 \forall s \leq t\}}_{\text{clean control: no history through } t}, \quad h < 0. \quad (4)$$

The washout guarantees an untreated history only over $[-120, -1]$, so at horizons $h < -120$ an event is not excluded merely because $t + h$ falls within an earlier episode of the same product. The coefficients at these distant horizons, which the figures in section 4.5 report back to $h = -500$, may therefore fall within, or remain affected by, an earlier treatment spell; this paper acknowledges that they cannot be guaranteed to be free of earlier treatment and should not be read as clean pre-treatment placebos. Guaranteeing an untreated history at such a horizon would require the washout to lengthen with the horizon: a coefficient at $h < -120$ is clean only if the product was untreated over $[h - 120, -1]$, not merely over $[-120, -1]$. Imposing this longer condition on every tenth coefficient from $h = -500$ to $h = -130$ leaves fewer events (six or seven vegetable and five or six fruit events, against nine and eight) and does not reduce the number of coefficients significant at the 5% level; with so few events, these estimates are imprecise. The pre-trend evidence on which identification rests is therefore confined to the window $[-120, -1]$ tested in section 4.4.

The key regressor ΔD_{it} equals one only on the day product i enters treatment ($D_{i,t-1} = 0$ & $D_{it} = 1$). It is an onset indicator, not a treatment level: the regression compares the price path following a fresh designation with the contemporaneous paths of clean controls, while duration of treatment is handled by the sample restriction, not by the regressor. For $h \geq 0$, β^h therefore measures the average log price gap h days after a clean onset among events still under treatment; for $h < 0$ it is the pre-onset price gap underlying the pre-trend diagnostics of section 4.4 below.

The treatment-intensity (continuous) specification scales the onset indicator by the product-specific, pre-determined intensity G_i , in %p:

¹⁰ The $j = 0$ component is already implied by $\Delta D_{it} = 1$.

$$\tilde{\Delta}^h \ln P_{it} = \beta^h (G_i \cdot D_i \cdot \Delta D_{it}) + T_t + X_{it} + \varepsilon_{it}^h, \quad \text{sample } \mathcal{S}_{\text{base}}^h, \quad (5)$$

so that β^h is expressed per 1%p of tariff reduction. Under the conditional parallel-trends assumption, β^h is a weighted combination of the effects of the individual onsets, each expressed per 1%p of its own tariff reduction (Callaway et al., 2024); it should therefore be interpreted as an average effect by treatment intensity rather than as a homogeneous marginal effect, and the per-1%p magnitudes reported below should be read in this way. Reading β^h as the effect of one more percentage point of reduction would further require that the effect of an onset be proportional to G_i and that the products with smaller reductions show how those with larger reductions would have responded to a smaller one, which pre-onset data cannot test. For each onset, G_i is the difference between the applied tariff rate and the Quota Tariff rate, the tariff reduction the designation delivers.¹¹

The resulting intensities, with the applied tariff rate and the Quota Tariff rate from which each is computed, are reported for all eleven ever-treated products in table 2; they run from 1.5%p for kiwifruit to 69.9%p for onion.

Table 2: Quota Tariff Treatment Intensity

Product	Applied Tariff Rate (%)	Quota Tariff Rate (%)	Treatment Intensity (%p)
Napa cabbage	27.0	0	27.0
Cabbage	26.9	0	26.9
Radish	29.4	0	29.4
Carrot	28.7	0	28.7
Onion	79.9	10	69.9
Green onion	27.0	0	27.0
Mango	15.7	0	15.7
Kiwifruit	6.5	5	1.5
Avocado	8.0	0	8.0
Pineapple	29.6	0	29.6
Banana	24.5	0	24.5

The impulse-response specifications estimate the same equations (2) and (5). Only the sample changes: the stay-on condition is removed from the treated side, and the

¹¹ Products with multiple onsets receive the average across their onsets, a single time-invariant scalar per product. Averaging over the pre-onset year smooths short-term fluctuations and keeps G_i pre-determined: rates realized after the onset, which could respond to the treatment itself, never enter. Section 3.4 details the applied tariff rate.

control side is unchanged:

$$\begin{aligned} \mathcal{S}_{\text{imp}}^h = & \underbrace{\{(i, t): \Delta D_{it} = 1, D_{i,t-s} = 0 \forall s \in [1, 120]\}}_{\text{clean treated: fresh entry, subsequent path unrestricted}} \\ & \cup \underbrace{\{(i, t): D_{is} = 0 \forall s \leq t, D_{i,t+j} = 0 \forall j \in [1, h]\}}_{\text{clean control: as in (3)}}, \quad h \geq 0, \end{aligned} \quad (6)$$

with coefficients denoted β_{imp}^h . For $h < 0$, the future no-treatment condition on the control side, $D_{i,t+j} = 0 \forall j \in [1, h]$, is dropped, as in the persistent-treatment specification; the treated-side condition remains unchanged, since it contains no forward-looking element to begin with:

$$\begin{aligned} \mathcal{S}_{\text{imp}}^h = & \underbrace{\{(i, t): \Delta D_{it} = 1, D_{i,t-s} = 0 \forall s \in [1, 120]\}}_{\text{clean treated: unchanged}} \\ & \cup \underbrace{\{(i, t): D_{is} = 0 \forall s \leq t\}}_{\text{clean control: no history through } t}, \quad h < 0. \end{aligned} \quad (7)$$

This coincides exactly with the persistent-treatment sample in equation (4). At horizons $h < 0$, the two specifications therefore share the same sample and the same regression, which makes their pre-event coefficient paths identical; this coincidence extends to $h = 0$ as well.

The intensity scaling in equation (5) is the one element that goes beyond the binary framework of Dube et al. (2025), and its interpretation follows the dose framework of de Chaisemartin and D'Haultfœuille (2026).

Standard errors are clustered by product throughout, and the pointwise 95% confidence bands in the figures use the normal critical value. Two caveats are in order. First, the number of *treated* clusters is small (six vegetable and five fruit products), and cluster-robust intervals therefore deserve caution (MacKinnon and Webb, 2018); section 5.4 examines the issue and the alternatives. Second, in the persistent-treatment specification the treated set itself thins as h grows (section 4.2). Both cautions inform the reading of the results.

The covariate vector X_{it} differs between the two groups of ever-treated products, each estimated in a separate regression. The two comparison pools share a common base, the 30 never-treated products, supplemented in each case with the pre-onset

observations of the group's own ever-treated products.¹² For the vegetable group,

$$X_{it}^{\text{veg}} = A_{it} + I_i \cdot O_t + I_i \cdot W1_{it} + I_i \cdot W2_{it} + R_{it} + Q_{it},$$

where A_{it} is the applied tariff rate (the tariff that would prevail absent the Quota Tariff; section 3.4), O_t is the West Texas Intermediate crude oil price, deflated by the U.S. consumer price index and divided by the exchange rate in KRW per USD, interacted with product dummies I_i to allow heterogeneous transport-cost channels, and $W1_{it}$ and $W2_{it}$ are climate covariates in the spirit of Roberts and Schlenker (2013): averages of daily temperature, humidity, precipitation, and sunshine over $[t - 100, t]$ ($W1_{it}$), and the same averages one year earlier (over $[t - 465, t - 365]$, $W2_{it}$), each interacted with product dummies. In that framework, current climate shifts supply directly, while the previous year's climate shapes planting decisions and hence current-year supply. R_{it} and Q_{it} collect the government stockpile-release and WTO Market Access Quota covariates of section 3.6. For the fruit group the climate blocks are omitted:

$$X_{it}^{\text{fruit}} = A_{it} + I_i \cdot O_t + R_{it} + Q_{it}.$$

The ever-treated fruits are predominantly imported, so Korean weather is not the relevant supply shifter for them, whereas domestic weather drives both vegetable supply and the timing of vegetable designations. Empirically, fruit pre-event coefficients show no systematic run-up without the climate covariates, whereas the vegetable pre-event run-up documented in section 4.4 below is larger without them.

Identification rests on the two assumptions standard for LP-DiD: no anticipation, and parallel trends of treated products relative to clean controls, conditional on X_{it} , the calendar-day effects, and product fixed effects.¹³ Anticipation has little room to operate, since some designations take effect with no advance legislative notice at all, notice approaching one month was given in just a few cases, and enforcement typically begins

¹² Treated Group 1 comprises the six leafy and root vegetables, and Treated Group 2 consists of the five fruits listed in section 3.1; when one group is estimated, the other group's products leave the sample. At far horizons of the impulse-response specification the estimates of the two groups have opposite signs (section 4.5), so a pooled coefficient would average a negative fruit estimate with a positive vegetable estimate.

¹³ Product fixed effects are absent from equations (2) and (5) by construction rather than omission: the dependent variable of equation (1) already subtracts each product's own pre-event mean.

immediately or within one to five days of promulgation (section 3.3). The pre-event coefficients over $[-120, -1]$ reported in section 4.5 bear directly on both assumptions. The LP-DiD results can be interpreted causally only if the conditional parallel-trends assumption holds, so the next subsection takes it to the data.

4.4 Pre-Designation Price Dynamics

The Quota Tariff is, by statutory design, aimed at products whose prices have surged, which may place treated products on price paths unlike those of the comparison pool precisely when treatment begins. The concern is an Ashenfelter dip in reverse (Ashenfelter, 1978). If designations are triggered by rising prices, a later decline occurring for reasons unrelated to the tariff cut could be mistaken for an effect of the policy. This subsection documents how retail prices move around designations and examines how much of the vegetable run-up the climate covariates absorb.

Figure 1: Retail Prices around Quota Tariff Onsets in Event Time

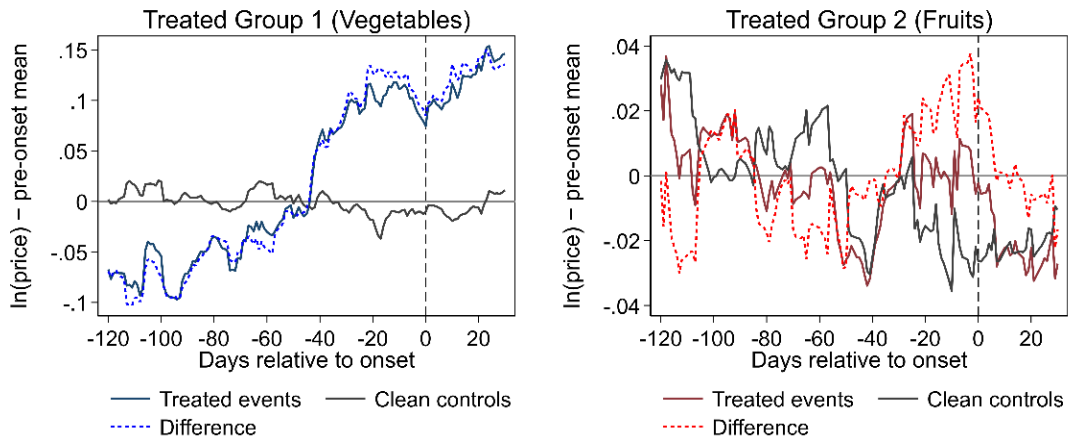


Figure 1 aligns every designation event on its onset date and traces average retail prices in event time, in the metric the estimation uses: the log retail price minus its own average over the 120 days preceding onset.¹⁴ The same transformation is applied to the

¹⁴ The events are those that enter the estimation, namely onsets preceded by at least 120 treatment-free days (section 4.2): nine vegetable onsets and eight fruit onsets. By construction each series averages to zero over its own 120-day pre-onset window, here and in figure 2; the diagnostic content lies in the

30 never-treated products on the same calendar dates, which lets market-wide movements enter both curves symmetrically. For vegetables the pattern is stark: prices climb from about 7 log points below the never-treated comparison four months before designation to 9–14 log points above it in the final three weeks, and the climb appears in eight of the nine events. For fruits there is no counterpart, the two curves staying within about 4 log points of each other throughout.

Whether this movement threatens identification depends on whether it survives the estimation’s conditioning. Figure 2 reports event-study coefficients at pre-onset horizons $h \in [-120, -1]$ from the treatment-status specification, estimated with and without the climate blocks on a common sample. Without the climate covariates the vegetable path reproduces that run-up in coefficient form: a significantly negative cluster deep in the pre-onset window, then significantly positive coefficients on the 39 consecutive days preceding onset, between +5 and +12 log points. With the climate blocks, adopted on supply-shifter grounds (Roberts and Schlenker, 2013), the same path moves toward zero, and none of its coefficients in the 37 days preceding onset is significant. The difference between the two vegetable paths, tested every 15 days from $h = -120$ to $h = -30$ with a cluster-jackknife standard error, is significant at $h = -75$ but not on average across these horizons. For the fruit group the two versions are indistinguishable; adding the climate blocks yields 11 significant positive coefficients, against 2 of 120 without them, which supports leaving the climate blocks out of the fruit specification.

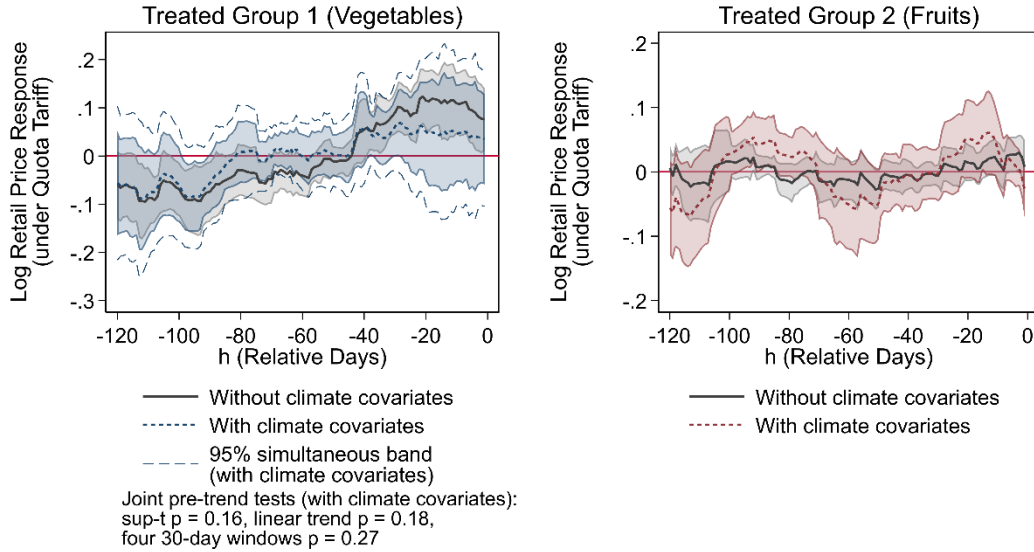
The parallel-trends assumption of section 4.3 is *conditional on the covariates*, and the climate covariates reduce the systematic pre-designation differences between the vegetable group and the comparison group without eliminating them. With the climate covariates the vegetable path is flatter but not flat: the negative coefficients over $h \in [-99, -91]$ remain individually significant (figure 2). These covariates are averages from a single weather station and therefore capture conditions in the main producing regions only imperfectly, and they cannot rule out unobserved shocks that affect both the designation decision and subsequent prices.

Because some pre-onset coefficients are individually significant, three joint tests, with the climate covariates in the vegetable regressions, examine the 120 coefficients

path within the window, not its level.

over $h \in [-120, -1]$ together: a 95% simultaneous confidence band, which checks whether any coefficient differs from zero once all 120 are considered jointly; a test for a linear trend in the coefficients; and a Wald (F) test that the averages over the four 30-day windows are equal. Section S7 reports the three tests in full and explains why a Wald test of all 120 coefficients cannot be computed with product-clustered standard errors. None of the three rejects at the 5% level in either group, in the treatment-status and the treatment-intensity specifications alike (table S7), and even over $h \in [-99, -91]$, where the vegetable coefficients are individually significant, the simultaneous band in figure 2 includes zero. These tests have limited power, so they show only that no violation is detectable at this resolution, not that the parallel-trends assumption holds. The remaining pre-onset differences therefore concern the vegetables: the fruit estimates, on which the main findings rest, are obtained without the climate covariates and show no pre-onset run-up, and for vegetables the paper claims no price-reducing effect. Subject to the two caveats of section 4.3, the analysis turns to the estimates.

Figure 2: Pre-Onset Coefficients with and without Climate Covariates



4.5 Graphical Results

Figures 3, 4, 5, and 6 present the four sets of estimates: the treatment-status specification

in its persistent-treatment and impulse-response forms, then the treatment-intensity specification in the same order. In all four, solid blue lines denote Treated Group 1 (vegetables), dashed red lines denote Treated Group 2 (fruits), and the shading indicates 95% confidence intervals (clustered by product). The treatment-intensity estimates are the main results of this study: a Quota Tariff onset lowers the tariff rate by 1.5%p to 69.9%p depending on the product, and the intensity specification accounts for that depth, whereas the treatment-status specification captures the adoption status regardless of the depth of the cut.

Figure 3: Treatment Status (Binary), Persistent-Treatment Specification

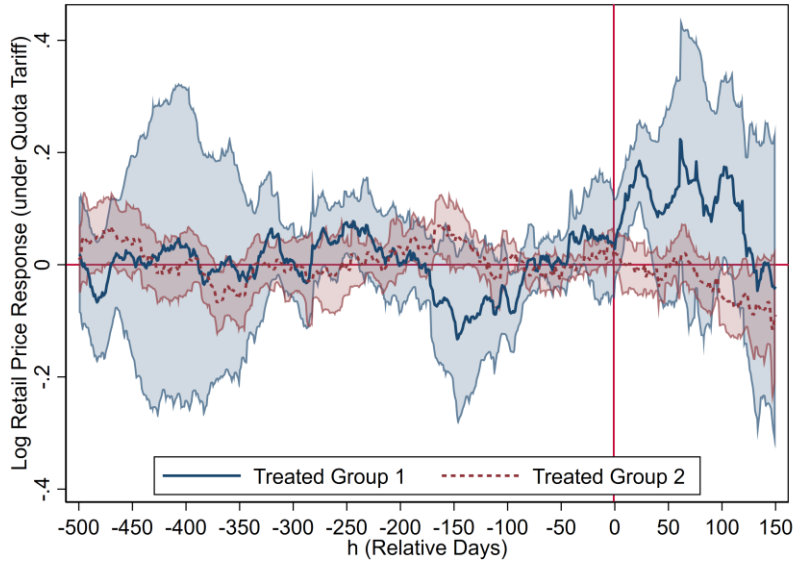


Figure 3 shows the treatment-status estimates under persistent treatment. Over $h \in [-120, -1]$, the joint tests of section 4.4 detect no violation of parallel pre-trends in either group. For vegetables (Treated Group 1), 10 of the 120 pre-event coefficients are significant at the 5% level, modestly above the number expected by chance, but they sit in a single nine-day negative cluster at $h \in [-99, -91]$ and one isolated positive day ($h = -38$), none within the 37 days preceding adoption. For fruits (Treated Group 2), only 5 of 120 are significant, below the number expected by chance, and all five form a short positive run at $h \in [-7, -3]$, too small and brief to disturb a reference level averaged over 120 pre-event days (equation (1)).

Post-onset the two groups diverge. Vegetable prices do not fall: the coefficients are

never significantly negative, turn significantly positive in the first weeks after adoption (over $h \in [7, 34]$), and are significant again only briefly through $h = 150$. Those early positives should not be read as the tariff cut raising prices: if the cut were the cause they would be larger where cuts are deeper, yet in the intensity specification below they are not significant.

Fruit prices, in contrast, begin to fall about three months after adoption: significant negative pockets open over $h \in [90, 131]$ (about -5.7 log points at $h = 100$) and recur through $h \approx 200$ in the replication files.¹⁵ Treating shallow-cut events such as kiwifruit (1.5%p) on the same footing as deep-cut events such as banana (about 25%p) makes the fruit response hard to estimate precisely; the treatment-intensity specification reveals it far more clearly.

Figure 4: Treatment Status (Binary), Impulse-Response Specification

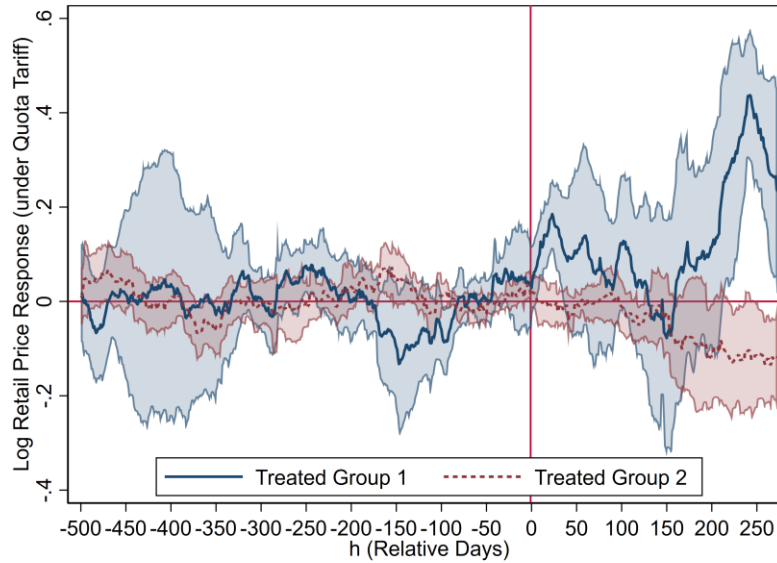
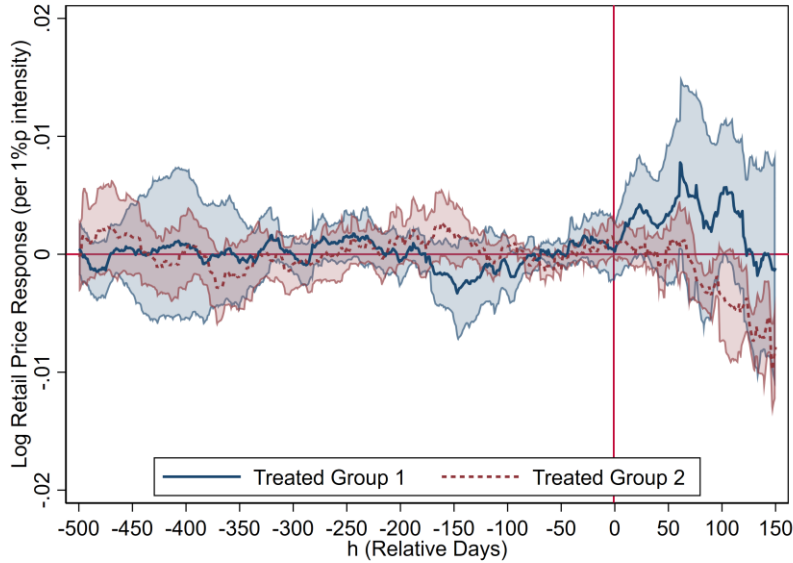


Figure 4 gives the impulse-response form of the treatment-status specification, the total effect of initiation, tracked to $h = 273$ with no attrition of treated products. For fruits (Treated Group 2), the initiation effect turns significantly negative in pockets from $h = 118$ onward and settles into a sustained decline at far horizons: -11.5 log points at $h = 250$, remaining there through $h = 273$, a decline of roughly 11%. These

¹⁵ However, the estimate at $h = 150$ (-8.95 log points) is imprecise.

coefficients do not by themselves show that the decline lasts after the designations end. Rather, they measure outcomes after policy initiation under the realized subsequent treatment paths. At $h = 250$, five of the eight fruit events are under treatment; the other three all began on the same day (table S6 in section S6), so the estimates at this horizon say little about whether the decline lasts after the designations end. For vegetables, apart from the early positive coefficients already seen under persistent treatment,¹⁶ the estimates are insignificant until about $h = 213$ and then turn sharply positive (+39 log points at $h = 250$). This far-horizon vegetable estimate should not be interpreted causally. It is driven by the four events initiated in 2024: napa cabbage, cabbage and carrot in May, and radish in July. Their 200–270-day windows fall between November 2024 and March 2025, when the kimchi-vegetable markets, radish and napa cabbage above all, stood far above their pre-designation levels in the wake of the autumn-2024 supply disruptions. These far-horizon vegetable coefficients pick up a supply disturbance specific to these products and are reported for transparency but excluded from the causal conclusions.

Figure 5: Treatment Intensity (per 1%p), Persistent-Treatment Specification



¹⁶ The two samples are mechanically identical at short horizons, before any spell has ended.

Figure 5 turns to the treatment-intensity specification, the persistent-treatment response per 1%p of the tariff reduction a Quota Tariff onset induces. For vegetables (Treated Group 1) none of the 120 pre-event coefficients is significant, and through $h = 150$ the only significant response is a two-day positive blip at $h \in [61, 62]$. Under ongoing treatment, no decline in vegetable prices per unit of tariff reduction is detectable. For fruits (Treated Group 2), 17 of 120 pre-event coefficients are significant, above the chance rate, and this pre-event diagnostic warrants scrutiny. The significant days fall in two clusters, one positive and one negative, both seven weeks or more before adoption, with none in the 49 days preceding the onset. Since adjacent coefficients are serially correlated, clusters like these can arise; near the Quota Tariff onset there is no systematic pattern of significant coefficients.

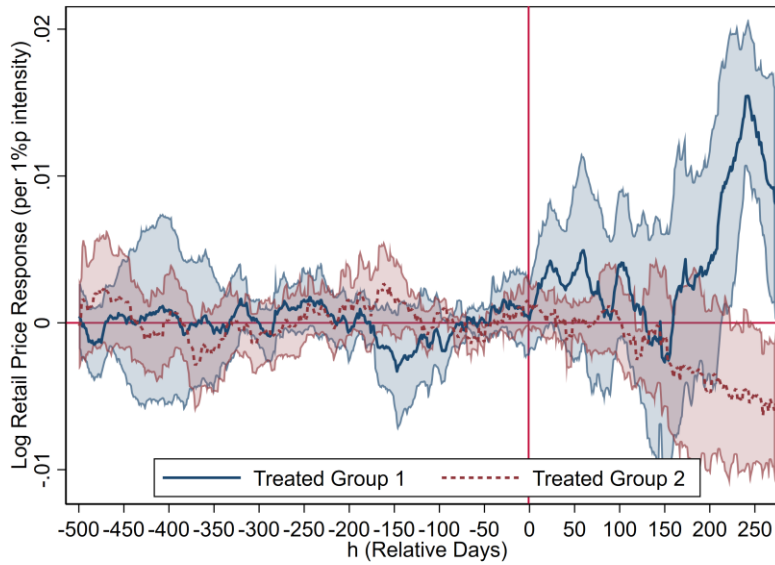
Post-onset the fruit response hovers near zero for the first ten weeks and is almost always negative thereafter. It is significant at $h \in [76, 96]$ and reaches -0.786 at the plotted window's edge, $h = 150$ (in log points per 1%p, i.e. the coefficient $\times 100$). Beyond that window the fruit treated set is unchanged through $h = 270$, and the replication files show the significant band continuing over $h \in [112, 209]$ and reaching -1.5 per 1%p at $h = 182$. The graphs stop at $h = 150$ only because of the attrition of treated events in the vegetable group. This significant fall in fruit retail prices between 90 and 210 days after onset is the core causal finding of the paper, and the sign of the response is robust across all four specifications; section 5.4 returns to mid-horizon magnitudes that exceed what full pass-through would imply.

Figure 6 shows the impulse responses of the treatment-intensity specification. For fruits (Treated Group 2), they repeat the pattern of their treatment-status counterparts: significant negative pockets from $h = 119$ onward, then a sustained significant decline at far horizons, -0.49% per 1%p at $h = 250$ and -0.55% at $h = 273$. For vegetables (Treated Group 1) the estimates are insignificant until about $h = 214$ and then jump abruptly positive ($+1.39\%$ per 1%p at $h = 250$), subject to the caution noted above for the treatment-status impulse responses.

Two diagnostics complete the review. The first is the reverse Ashenfelter dip of section 4.4, which three features argue against: the pre-event coefficients are

insignificant in the window adjacent to adoption in all four specifications;¹⁷ equation (1) fixes the reference level at a 120-day average that a surge in the last few days barely moves, while the calendar-day fixed effects absorb market-wide surges; and the vegetable climate covariates absorb part of the supply shocks behind both price surges and designations (figure 2). A classic negative dip is likewise absent. One case that cannot be fully excluded is anticipation happening to offset a dip exactly, but such a coincidental offset is highly unlikely because, as described earlier, anticipation has little room to operate.

Figure 6: Treatment Intensity (per 1%p), Impulse-Response Specification



The second diagnostic is economic magnitude. Under full and immediate pass-through, a 1%p tariff reduction lowers the consumer price by approximately $100/(100 + \bar{\tau})\%$, where $\bar{\tau}$ is the prevailing tariff level; for the ever-treated fruit products the counterfactual applied tariff rate $\bar{\tau}$ ranges from about 6.5% to 29.6% (table 2), placing the full-pass-through benchmark between roughly 0.77% and 0.94% per 1%p. The estimated fruit responses (-0.78% per 1%p at $h = 150$ under ongoing treatment, and -0.49% to -0.63% per 1%p at $h \geq 250$ for initiations) therefore imply substantial, at some horizons near-complete, pass-through of the tariff reduction to

¹⁷ The exceptions are the five-day blip for fruits and the single positive day at $h = -38$ for vegetables, both discussed above.

consumers.¹⁸ The magnitude varies with horizon and specification, and these estimates bracket the plausible range rather than pin down a single pass-through rate.

4.6 LP-DiD Estimates at Selected Horizons

Table 3 condenses the four sets of estimates into regression form at horizons chosen by the design, not by the results. For the persistent-treatment specification, $h = 150$ is the truncation point of its graphs, a conservative margin below the compositional break at $h = 156$. The impulse-response specification is reported at the same horizon for comparability and at $h = 250$, roughly eight months after initiation, where its treated sets are still complete (nine vegetable and eight fruit events). Coefficients and standard errors are multiplied by 100 for readability.

The table restates the four figures in compact form. At $h = 150$ in the persistent-treatment specification, the only significant estimate is the fruit response per 1%p of intensity, -0.786 (in log points, i.e. the coefficient $\times 100$; panel A, column 4); the fruit estimate in the treatment-status specification is negative but insignificant (panel A, column 2), and both vegetable estimates (panel A, columns 1 and 3) are imprecise. At the same horizon the impulse-response estimates (panel B) are insignificant throughout. Contrasting panel B with panel C shows how long the effect of initiation takes to materialize: by $h = 250$ it emerges as significant fruit declines (-11.5 binary, panel C, column 2; -0.50 per 1%p, panel C, column 4). The significance of the fruit estimates thus depends on the estimand, the treatment definition and the horizon: three of the six are significant at the 5% level. As detailed above, the positive far-horizon vegetable estimates (panel C, columns 1 and 3) reflect the contemporaneous supply disturbance in the products designated in 2024 rather than causal effects.

Removing the product with the largest tariff reduction leaves the sign and 5% significance of every treatment-intensity estimate unchanged but changes the size of the fruit impulse-response estimates considerably, which indicates that the responses per

¹⁸ Computed as $(e^{\beta^h} - 1) \times 100$: -0.783% at $h = 150$ (persistent treatment) and between -0.487% ($h = 252$) and -0.629% ($h = 259$) over $h \in [250, 273]$ (impulse response). The benchmark treats the quota as non-binding within the window and abstracts from inventory and contracting lags, making it an upper bound on the speed, not the existence, of pass-through.

1%p of tariff reduction differ across products. The direction of the fruit decline therefore does not rest on the product with the largest tariff reduction, whereas the per-1%p magnitudes in table 3 should be read as average effects by treatment intensity (section 4.3), not as the effect of one more percentage point of tariff reduction.

Table 3: Local Projection Difference-in-Differences Estimates at Selected Horizons

	(1) Quota Tariff Treatment Status Group 1 (Vegetables)	(2) Group 2 (Fruits)	(3) Quota Tariff Treatment Intensity Group 1 (Vegetables)	(4) Group 2 (Fruits)
<i>Panel A: Persistent treatment, $h = 150$</i>				
Quota Tariff Shock	-3.894 (14.756)	-8.950 (6.166)	-0.117 (0.515)	-0.786*** (0.224)
Observations	33678	45026	33678	45026
R^2	0.594	0.159	0.594	0.159
Adjusted R^2	0.577	0.131	0.577	0.131
<i>Panel B: Initiation (impulse response), $h = 150$</i>				
Quota Tariff Shock	-7.770 (12.793)	-2.338 (4.400)	-0.268 (0.448)	-0.088 (0.247)
Observations	33682	45030	33682	45030
R^2	0.594	0.159	0.594	0.159
Adjusted R^2	0.577	0.131	0.577	0.131
<i>Panel C: Initiation (impulse response), $h = 250$</i>				
Quota Tariff Shock	38.813*** (7.518)	-11.519** (5.653)	1.379*** (0.268)	-0.495** (0.237)
Observations	30154	41530	30154	41530
R^2	0.703	0.191	0.703	0.191
Adjusted R^2	0.691	0.164	0.691	0.164

Coefficients and standard errors multiplied by 100 for readability; standard errors, in parentheses, clustered by product (36 clusters in columns 1 and 3, 35 clusters in columns 2 and 4).

Panel A conditions on treatment persisting through h ; panels B and C track initiations regardless of the subsequent path.

Treated clusters differ by panel because the stay-on condition of panel A removes events as h grows: panel A rests on 5 treated vegetable and 4 treated fruit products, panels B and C on 6 and 5, the full ever-treated sets.

Significance is based on the t distribution with degrees of freedom equal to the number of product clusters minus one.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

If the effect of an onset is not proportional to its tariff reduction, the product with the largest reduction can weigh heavily in the treatment-intensity coefficient (section 4.3). Removing, in each group, the ever-treated product with the largest tariff reduction (onion among the vegetables, pineapple among the fruits) leaves the sign and 5% significance of all six treatment-intensity estimates in the table unchanged (table S8 in section S8). The price responses after the pineapple onsets are weak relative to their

tariff reductions and pull the fruit impulse-response estimates toward zero, so these estimates become larger in absolute value without pineapple. The check does not bear on the fruit estimate under persistent treatment at $h = 150$, because both clean pineapple onsets leave that sample earlier.

Taken together, the causal analysis yields one policy-relevant effect and one clearly labeled null result. Quota Tariff designations on fruits significantly lower retail prices per 1%p of tariff reduction while the reduction is in force ($h = 150$), and retail prices are also significantly lower at $h = 250$ after initiation, when five of the eight fruit events are under treatment; designations on leafy and root vegetables, by contrast, show no detectable price-reducing effect at any interpretable horizon, under either estimand. The next section probes the robustness of these conclusions and interprets the difference between the fruit and vegetable results.

5 Robustness Check

Beyond the pre-onset tests of the conditional parallel-trends assumption in section 4.4, the causal interpretation of the estimates in section 4 rests on three further assumptions: that the effects of an earlier episode fade within the 120-day washout, that a designation does not move the prices of products in the comparison pool, and that the comparison pool and the ever-treated products are not exposed differently to price movements after onset. Section 5.1 re-estimates the specifications of table 3 with longer washout periods and on the first onset of each product, and the persistent-treatment specification on a fixed set of treated onsets. The fruit response at long horizons, at $h = 250$ in the impulse-response specification, rests largely on the second designations of mango, pineapple and banana and is not robust to a longer washout, which qualifies the long-horizon fruit decline reported in sections 4.5 and 4.6; neither the fruit decline under persistent treatment nor the positive vegetable coefficients in the first weeks after onset arise from the changing composition of the treated set. Section 5.2 removes candidate substitutes and complements from the comparison pool, which leaves the signs and 5% significance of all 12 estimates in table 3 unchanged. Section 5.3 examines what explains the difference between the fruit and vegetable results and, among its checks of the fruit decline, addresses the third assumption by restricting the never-treated part of

the fruit comparison pool to import-dependent products, which makes the fruit estimates more negative. Table S9 in section S9 summarizes how the main estimates change under these checks. Section 5.4 then turns to inference with few treated clusters, which does not address these assumptions.

5.1 The Washout Condition

If the effects of an earlier episode last longer than 120 days, the prices after an onset that passes the washout may still reflect them. The washout length determines which onsets enter as clean events: a longer washout also removes onsets that follow a previous episode by more than 120 days but within the longer washout period. Table S10 in section S10 re-estimates the six regressions of table 3 for each group with washout periods of 180 and 240 days and on the first onset of each product only. The vegetable estimates keep their signs and 5% significance wherever they can be assessed. The fruit estimates are unchanged with a 180-day washout, but the long-horizon estimates, at $h = 250$ in the impulse-response specification, become smaller in absolute value and insignificant with a 240-day washout and in the first-onset sample (table S9); in the persistent-treatment specification the fruit estimates cannot be assessed beyond a 180-day washout, since three or fewer treated products remain. The onsets removed are the second designations of mango and pineapple and, in the first-onset sample, also that of banana. The first-onset sample carries little information on its own, because three of its five fruit onsets fall on the same day.

The treated set of the persistent-treatment specification changes not only with the washout but also across horizons before $h = 150$: as table S6 in section S6 shows, onsets leave the sample when their spells end, at $h = 45, 52, 61, 104$ and 120 , so the estimates up to $h = 150$ are not based on the same set of events. To check whether this change drives the results, the persistent-treatment specification is re-estimated on a fixed set of treated onsets: the five vegetable and four fruit onsets that remained under the Quota Tariff for at least the first 150 days. These onsets enter every horizon from $h = 0$ to $h = 150$, and the control side follows the baseline rules (table S11 in section S10). For fruits the two designs coincide from $h = 60$, because only these four onsets remain from that horizon; at the shorter horizons that were estimated, no fruit estimate is significant at the 5% level in either design. The fruit decline therefore does not arise

from the changing composition. For vegetables the two designs coincide from $h = 120$, and the positive coefficients in the first weeks after onset also appear in the fixed set, so they do not arise from the changing composition either.

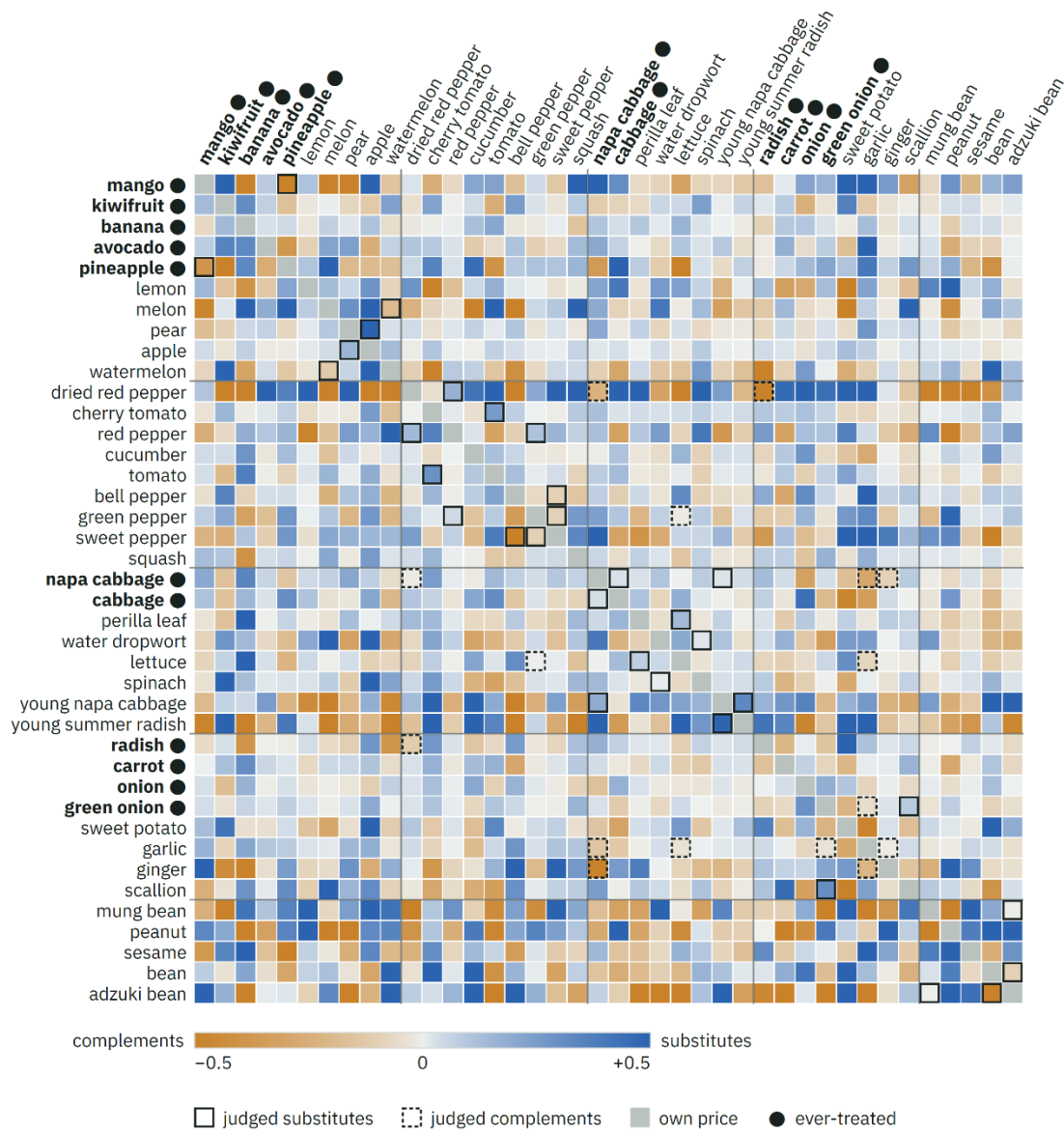
5.2 The Stable Unit Treatment Value Assumption

The main estimates assume SUTVA, which requires here that the price of a product in the comparison pool not respond to the designation of another product; this section relaxes the assumption for the products most likely to respond. If a designation lowers the price of the treated product and, through consumer substitution, also that of a substitute in the comparison pool, the treated–control difference is compressed and the estimate is biased toward zero; spillovers to a complement bias it in the opposite direction. Close substitutes and complements of an ever-treated product may therefore not be appropriate controls for it, and the sensitivity of the estimates to such spillovers is assessed by removing these products from the comparison pool. To select candidates for removal, compensated cross-price elasticities between the products are estimated with the almost ideal demand system of Deaton and Muellbauer (1980), from weekly point-of-sale data of large retailers and supermarket chains, since the system needs quantities sold as well as prices. Figure 7 shows the estimated elasticities: each cell gives the response of the quantity demanded of the row product to the price of the column product.¹⁹ The figure outlines the pairs of products that one would expect to be substitutes or complements, and for most of these pairs the estimated sign agrees with that expectation. The products whose elasticities with respect to the price of an ever-treated product k are most clearly estimated are taken as candidate substitutes and complements of k , and the exclusion specifications remove them from the comparisons around each designation of k ; section S11 describes the demand system, the selection rule and the exclusion window and summarizes the comparison with the expected signs. Young napa cabbage, scallion and young summer radish, closely related to napa cabbage, green onion and radish, respectively, are also removed from every vegetable comparison in these specifications, whatever their estimated elasticities; table 3 and the figures in section 4 keep them in the comparison pool. Table S12 in section S11 reports

¹⁹ An interactive version of figure 7 is available at https://jayjeo.github.io/github_TRQ/aids-map/.

the estimates obtained when substitutes and complements are removed together and when each is removed separately. With both removed, at least 21 control products remain on every comparison date in both groups from $h = 0$ through the horizons of table 3. In all three cases, the 12 estimates in that table keep their signs and 5% significance, so the conclusions drawn from that table do not depend on keeping these candidates in the comparison pool.

Figure 7: Compensated Cross-Price Elasticities between the Products



5.3 Import Dependence

This section examines what explains the difference between the fruit and vegetable results. The estimates identify a decline in retail prices for the fruit group and detect no price-reducing effect for the vegetable group. Import dependence is far higher for the ever-treated fruits than for most ever-treated vegetables (table S4), which is consistent with this difference in results. However, import dependence differs between the two product groups, as do other characteristics of fruits and vegetables, such as perishability and their distribution systems, so a comparison of the two groups cannot tell whether import dependence or one of these other characteristics causes the difference in results. The rest of this section shows that the share of supply that a designation affects directly is also consistent with the difference in results but, like import dependence, differs between the two product groups, and that the vegetable data cannot test the import-dependence mechanism within the group. For the fruit group, the import-dependence mechanism therefore remains an untested hypothesis. The section also shows that the fruit decline at $h = 150$ under persistent treatment survives checks against exchange-rate movements and product-specific trends, is already present at the wholesale stage and does not depend on a comparison with products of low import dependence. It then shows that a direct test does not establish that the two groups respond differently, and it compares the fiscal cost of the designations.

The Quota Tariff lowers the tariff only on imports, so its effect can reach retail prices only through the imports whose tariff a designation reduces, and the share of supply that enters on the tariff lines for fresh produce covered by a designation (policy exposure) differs sharply between the two groups. Mango, banana, avocado and pineapple, which have no domestic production data, are recorded in the retail price data only as imported varieties (table S13 in section S12), so for these four tropical fruits a designation lowers the tariff on the very goods that make up their retail prices. None of the surveyed varieties of the six ever-treated vegetables is recorded as imported, but their retail prices are prices of both domestic and imported produce. A designation can reach the domestic produce only through competition with imports, and for every vegetable except carrot the imports that entered on the tariff lines for fresh produce covered by a designation were small relative to supply in 2021, before the first designation (table S14 in section S12). This difference in the share of supply is

consistent with the decline identified in the fruit group and the absence of a detected price-reducing effect in the vegetable group, but the share, like import dependence, differs between the two product groups, so the estimates cannot test whether it is the cause of these results.

One way to test the import-dependence mechanism while holding the product group fixed would be to compare vegetables that differ in their 2021 policy exposure (table S14). Policy exposure can be far smaller than import dependence: the import dependence of napa cabbage is 21.4% (table S4), but its imports are almost entirely kimchi, and no kimchi tariff line was designated between 2015 and 2025, so the policy exposure of napa cabbage is effectively zero. Among the ever-treated vegetables only carrot has a meaningful policy exposure, and carrot has a single clean onset. With one onset, the minimum detectable effect is about 33 to 54 log points (section S12), whereas the tariff reduction would lower carrot prices by 0 to 13 log points under different pass-through assumptions. The vegetable estimates therefore cannot test the import-dependence mechanism.

Policy exposure also yields, without using any price data, a predicted price change for each product: the change that full pass-through of the tariff reduction would imply, computed from pre-designation customs and production data (table S14). The predicted declines are 7.7 to 26.2 log points for the four tropical fruits, but at most 1.3 for five of the six vegetables and for kiwifruit, whose tariff was reduced by only 1.5%p; for carrot the prediction is 12.5, too small to detect with its single onset. A decline is identified for the fruits, and no price-reducing effect is detected for the vegetables, a pattern consistent with these predictions. However, because policy exposure, like import dependence, differs between the two product groups, this correspondence is not a test of the mechanism.

The fruit decline could have causes other than the designations. Because the ever-treated fruits are mostly imported, their prices could have fallen with the exchange rate; they could also have followed their own trends, fruit prices could have fallen in general, or retailers could have narrowed their margins. Checks of these four explanations do not support any of them. First, exchange-rate movements do not explain the decline: at $h = 150$ under persistent treatment, the treatment-intensity estimate remains negative and significant when the fruit regressions add product fixed effects and product-specific

coefficients on the level of the KRW/USD exchange rate. Second, the decline does not reflect trends in the prices of the ever-treated fruits: the same estimate remains negative and significant when the regressions add only product-specific linear time trends, and the decline does not reappear when the treatment paths of these fruits are shifted 730 days earlier, to placebo dates. Third, the decline is not a fall in fruit prices in general: when the treatment paths are assigned to the five never-treated fruits, these fruits do not show the decline. The two placebo checks use the treatment-status specification, because the assigned paths carry no tariff reduction. Fourth, the decline does not come from narrower retail margins: in the treatment-intensity specification at $h = 150$, wholesale prices of the ever-treated fruits also fell significantly (-1.09 log points per 1%p, $p = 0.002$), and the retail-to-wholesale margin did not change significantly, so the decline is already present at the wholesale stage. The wholesale estimate, however, cannot separate the tariff reduction from other shocks before the retail stage, such as exchange-rate movements. Table S15 in section S12 gives the details: panel A reports the exchange-rate and trend checks, panel B reports the two placebo checks, and panel C reports the wholesale and margin results.

Table S4 in section S1 shows that the comparison pool is not balanced in import dependence: the ever-treated vegetables are less import-dependent than the products in the pool, and the ever-treated fruits are far more import-dependent than these products. A difference in import dependence violates the parallel-trends assumption only if it gives the ever-treated products and the pool different price trends, and before onset the joint tests of section 4.4 detect no such difference; after onset, however, the fruits could be more exposed than the pool to exchange-rate or world-price movements. To check whether the difference in import dependence drives the fruit estimates, the fruit regressions are re-estimated with the never-treated part of the comparison pool restricted to the products whose pre-treatment import dependence is at least 50% (dried red pepper, mung bean, peanut, sesame, bean and adzuki bean) and to lemon (table S16 in section S12); the import dependence of lemon is set to 100% in table S4 because no domestic production is recorded. The fruit estimates become more negative (table S9), and the treatment-status estimate under persistent treatment at $h = 150$ becomes significant at the 5% level, so the fruit estimates do not depend on a comparison with products of low import dependence. The fruit regressions with this restricted pool have only 12 product clusters, and most of the never-treated products in the pool are subject

to state trading and Market Access Quotas, which partly insulate their domestic prices from world prices. The check is not applied to vegetables, whose comparison pool is already more import-dependent than the ever-treated vegetables, so for them the evidence on this imbalance rests on the pre-onset tests.

The contrast in significance between the fruit and vegetable estimates is not by itself evidence that their responses differ. Table S17 in section S12 reports a direct test of whether the fruit and vegetable responses differ. At $h = 150$ the difference is not significant in any of the four specifications, mainly because the vegetable estimates at this horizon are imprecise. At $h = 250$ in the impulse-response specification the difference is significant, but it arises from the positive vegetable estimates that section 4.5 attributes to the supply disturbance in the products designated in 2024 and excludes from the causal conclusions. The test therefore does not show that the two groups respond differently; the heterogeneity described above consists in a decline identified for fruits and no price-reducing effect detected for vegetables.

The designations also differ in their fiscal cost, the tariff revenue forgone on imports that enter at the Quota Tariff rate. Assuming that the same quantities would have been imported without the designations, the designations on the tariff lines for the fresh produce of the six ever-treated vegetables between 2021 and March 2025 cost about USD 33 to 54 million, more than four fifths of it on carrot and onion, and those of the five ever-treated fruits about USD 176 million (section S12). Onion's small policy exposure does not imply a small fiscal cost: its exposure is measured with 2021 imports, which were small relative to supply, but within its designated periods in 2022 and 2023 onion imports reached 89,450 metric tons. For the fruits this cost came with an identified price reduction, whereas the vegetable designations have no identified effect on prices.

5.4 Inference with Few Treated Clusters

The number of treated clusters is set by the policy itself: over the sample period the Quota Tariff reached six vegetable and five fruit products in the sample, and the stay-on condition leaves five and four of them in the persistent-treatment regressions at $h = 150$. With so few treated clusters, cluster-robust t -statistics tend to over-reject (Conley and Taber, 2011; MacKinnon and Webb, 2020), while the restricted wild cluster bootstrap under-rejects in this configuration (MacKinnon et al., 2023). Randomization,

placebo, rescaling and wild-bootstrap alternatives restore validity only by imposing homogeneity assumptions that the present setting violates, and all were developed for a binary treatment that turns on once and stays on; section S13 reviews them and shows why none applies here.

The analysis therefore reports conventional product-clustered intervals and reads them in light of the over-rejection of cluster-robust t -statistics. Over-rejection is a bias toward finding effects, so the conclusion that no price-reducing effect is detectable for vegetables is reached against that bias: even the conventional tests find no significantly negative vegetable coefficient after onset in figures 3–6. The significant positive vegetable coefficients in the first weeks after onset and at far impulse-response horizons are not read as effects of the tariff reduction, for the reasons given in section 4.5. At the horizons of table 3 the magnitude of the fruit effect lies at or below the full-pass-through benchmark of section 4.5, but larger mid-horizon estimates, such as -1.5 per 1%p at $h = 182$, exceed that benchmark and therefore cannot be attributed to the tariff reduction alone. Following the reporting guidance of MacKinnon et al. (2023), the number of clusters and the treated-cluster counts are reported with the estimates in table 3, and the event-level composition of the treated sets, together with the horizons at which events exit them, is disclosed in section S6.

6 Conclusion and Policy Implications

This paper identifies the causal effect of Korea’s Quota Tariff on agricultural retail prices. Because a designation lapses when its announced period ends or its quota is exhausted, and the same product can be designated again, the treatment is non-absorbing. Methodologically, the paper adapts the LP-DiD framework of Dube et al. (2025), with its sample-selection rules for clean controls and clean treated units, and estimates the two effects that the framework defines: the effect while a designation is in force (persistent treatment) and the effect of a designation’s onset (impulse response).

Two findings follow. First, the results are heterogeneous: for leafy and root vegetables no price-reducing effect is detectable at any interpretable horizon under either estimand, whereas for fruits, prices fall by about 0.78% per 1%p of tariff

reduction at $h = 150$ while the reduction is in force, and by 0.49% to 0.63% per 1%p at $h \geq 250$ for initiations. Set against the 0.77% to 0.94% per 1%p implied by full and immediate pass-through, the fruit estimates indicate that a substantial share of the reduction reached consumers, at the shorter horizon almost all of it. Second, the effect for fruits emerges only after several months. The impulse-response estimates measure outcomes after onset under the realized subsequent treatment paths, and at $h = 250$ five of the eight fruit events are still under treatment, so these estimates do not by themselves show that the decline persists after the designations end.

The heterogeneity between the two groups consists in a decline identified for fruits and no price-reducing effect detected for vegetables; a direct test does not establish that their responses differ. Import dependence is far higher for the ever-treated fruits than for most ever-treated vegetables, which is consistent with the difference between the fruit and vegetable results. However, import dependence differs between the two product groups, as do other characteristics of fruits and vegetables, such as perishability and their distribution systems, so a comparison of the two groups cannot tell whether import dependence or one of these other characteristics causes the difference in results.

The designations also differ in their fiscal cost, the tariff revenue forgone on imports that enter at the Quota Tariff rate. Assuming that the same quantities would have been imported without the designations, the designations on the tariff lines for the fresh produce of the six ever-treated vegetables between 2021 and March 2025 cost about USD 33 to 54 million, more than four fifths of it on carrot and onion, and those of the five ever-treated fruits about USD 176 million. For the fruits this cost came with an identified price reduction, whereas the vegetable designations have no identified effect on prices.

References

- Amiti, M., Redding, S.J., Weinstein, D.E., 2019. The Impact of the 2018 Tariffs on Prices and Welfare. *Journal of Economic Perspectives* 33, 187–210. doi:[10.1257/jep.33.4.187](https://doi.org/10.1257/jep.33.4.187).
- Ashenfelter, O., 1978. Estimating the Effect of Training Programs on Earnings. *The Review of Economics and Statistics* 60, 47–57. doi:[10.2307/1924332](https://doi.org/10.2307/1924332).
- Bach, P., Klaassen, S., Kueck, J., Mattes, M., Spindler, M., 2025. Sensitivity analysis for treatment effects

- in Difference-in-Differences models using Riesz representation. [arXiv:2510.09064](#).
- Bellemare, M.F., Wichman, C.J., 2020. Elasticities and the Inverse Hyperbolic Sine Transformation. *Oxford Bulletin of Economics and Statistics* 82, 50–61. doi:[10.1111/obes.12325](#).
- Borusyak, K., Jaravel, X., Spiess, J., 2024. Revisiting Event-Study Designs: Robust and Efficient Estimation. *Review of Economic Studies* 91, 3253–3285. doi:[10.1093/restud/rdae007](#).
- Busch, A., Girardi, D., 2024. LPDID: Stata module implementing local projections difference-in-differences (LP-DiD) estimator. Statistical Software Components S459273, Boston College Department of Economics.
- Callaway, B., Goodman-Bacon, A., Sant’Anna, P.H.C., 2024. Difference-in-Differences with a Continuous Treatment. Working Paper 32117. National Bureau of Economic Research. Cambridge, MA. doi:[10.3386/w32117](#).
- Callaway, B., Sant’Anna, P.H.C., 2021. Difference-in-Differences with multiple time periods. *Journal of Econometrics* 225, 200–230. doi:[10.1016/j.jeconom.2020.12.001](#).
- Cavallo, A., Gopinath, G., Neiman, B., Tang, J., 2021. Tariff Pass-Through at the Border and at the Store: Evidence from US Trade Policy. *American Economic Review: Insights* 3, 19–34. doi:[10.1257/aeri.20190536](#).
- Cengiz, D., Dube, A., Lindner, A., Zipperer, B., 2019. The Effect of Minimum Wages on Low-Wage Jobs. *The Quarterly Journal of Economics* 134, 1405–1454. doi:[10.1093/qje/qjz014](#).
- Cleveland, R.B., Cleveland, W.S., McRae, J.E., Terpenning, I., 1990. STL: A seasonal-trend decomposition procedure based on loess. *Journal of Official Statistics* 6, 3–73.
- Conley, T.G., Taber, C.R., 2011. Inference with “Difference in Differences” with a Small Number of Policy Changes. *The Review of Economics and Statistics* 93, 113–125. doi:[10.1162/REST_a_00049](#).
- de Chaisemartin, C., D’Haultfœuille, X., 2020. Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review* 110, 2964–2996. doi:[10.1257/aer.20181169](#).
- de Chaisemartin, C., D’Haultfœuille, X., 2023. Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: A survey. *The Econometrics Journal* 26, C1–C30. doi:[10.1093/ectj/utac017](#).
- de Chaisemartin, C., D’Haultfœuille, X., 2026. Difference-in-Differences Estimators of Intertemporal Treatment Effects. *The Review of Economics and Statistics* 108, 863–880. doi:[10.1162/rest_a_01414](#).
- Deaton, A., Muellbauer, J., 1980. An Almost Ideal Demand System. *American Economic Review* 70, 312–326.
- Dube, A., Girardi, D., Jordà, Ò., Taylor, A.M., 2025. A Local Projections Approach to Difference-in-Differences. *Journal of Applied Econometrics* 40, 741–758. doi:[10.1002/jae.70000](#).
- Dube, A., Lindner, A., 2024. Minimum wages in the 21st century. *Handbook of Labor Economics* 5, 261–383. doi:[10.1016/bs.heslab.2024.11.004](#).

- Fajgelbaum, P.D., Goldberg, P.K., Kennedy, P.J., Khandelwal, A.K., 2020. The Return to Protectionism. *The Quarterly Journal of Economics* 135, 1–55. doi:[10.1093/qje/qjz036](https://doi.org/10.1093/qje/qjz036).
- Fajgelbaum, P.D., Khandelwal, A.K., 2026. Tariffs in 2025: Short-Run Impacts on the U.S. Economy. Working Paper 35064. National Bureau of Economic Research. doi:[10.2139/ssrn.6566859](https://doi.org/10.2139/ssrn.6566859).
- Flaen, A., Hortaçsu, A., Tintelnot, F., 2020. The Production Relocation and Price Effects of US Trade Policy: The Case of Washing Machines. *American Economic Review* 110, 2103–2127. doi:[10.1257/aer.20190611](https://doi.org/10.1257/aer.20190611).
- Gardner, J., 2022. Two-stage Differences in Differences. [arXiv:2207.05943](https://arxiv.org/abs/2207.05943).
- Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. *Journal of Econometrics* 225, 254–277. doi:[10.1016/j.jeconom.2021.03.014](https://doi.org/10.1016/j.jeconom.2021.03.014).
- Hillen, J., 2019. Market Integration and Market Efficiency under Seasonal Tariff Rate Quotas. *Journal of Agricultural Economics* 70, 859–873. doi:[10.1111/1477-9552.12355](https://doi.org/10.1111/1477-9552.12355).
- Jang, S.H., 2025. Quota Tariff Implementation: Current Status and Policy Improvement Agenda (할당관세 운용 현황과 개선과제). *National Assembly Budget Office (국회예산정책처)*.
- Levinsohn, J., 1993. Testing the Imports-as-Market-Discipline Hypothesis. *Journal of International Economics* 35, 1–22. doi:[10.1016/0022-1996\(93\)90002-F](https://doi.org/10.1016/0022-1996(93)90002-F).
- Livingston, A., Schaefer, K.A., Malone, T., 2025. Market Access Under a Dynamic Safeguard Mechanism: Evidence From the United States–Japan Trade Agreement. *Agricultural Economics* 56, 895–904. doi:[10.1111/agec.70039](https://doi.org/10.1111/agec.70039).
- Loginova, D., Portmann, M., Huber, M., 2021. Assessing the Effects of Seasonal Tariff-rate Quotas on Vegetable Prices in Switzerland. *Journal of Agricultural Economics* 72, 607–627. doi:[10.1111/1477-9552.12424](https://doi.org/10.1111/1477-9552.12424).
- MacKinnon, J.G., Nielsen, M.Ø., Webb, M.D., 2023. Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics* 232, 272–299. doi:[10.1016/j.jeconom.2022.04.001](https://doi.org/10.1016/j.jeconom.2022.04.001).
- MacKinnon, J.G., Webb, M.D., 2018. The wild bootstrap for few (treated) clusters. *The Econometrics Journal* 21, 114–135. doi:[10.1111/ectj.12107](https://doi.org/10.1111/ectj.12107).
- MacKinnon, J.G., Webb, M.D., 2020. Randomization inference for difference-in-differences with few treated clusters. *Journal of Econometrics* 218, 435–450. doi:[10.1016/j.jeconom.2020.04.024](https://doi.org/10.1016/j.jeconom.2020.04.024).
- Mollins, J., Lumb, R., 2024. Seasonal Adjustment of Weekly Data. Staff Discussion Paper 2024-17. Bank of Canada.
- Pierce, D.A., Grupe, M.R., Cleveland, W.P., 1984. Seasonal Adjustment of the Weekly Monetary Aggregates: A Model-Based Approach. *Journal of Business & Economic Statistics* 2, 260–270. doi:[10.1080/07350015.1984.10509393](https://doi.org/10.1080/07350015.1984.10509393).
- Roberts, M.J., Schlenker, W., 2013. Identifying Supply and Demand Elasticities of Agricultural Commodities: Implications for the US Ethanol Mandate. *American Economic Review* 103, 2265–2295. doi:[10.1257/aer.103.6.2265](https://doi.org/10.1257/aer.103.6.2265).

- Roth, J., Sant'Anna, P.H.C., Bilinski, A., Poe, J., 2023. What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics* 235, 2218–2244. doi:[10.1016/j.jeconom.2023.03.008](https://doi.org/10.1016/j.jeconom.2023.03.008).
- Schaefer, K.A., Wolf, C.A., 2025. Trade protection via tariff rate quota administration. *Food Policy* 131, 102801. doi:[10.1016/j.foodpol.2025.102801](https://doi.org/10.1016/j.foodpol.2025.102801).
- Song, Y.G., 2023. 2022 Quota Tariff implementation results and impact analysis (2022년도 할당관세 지원실적 및 효과분석). *Ministry of Economy and Finance (기획재정부)*.
- Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics* 225, 175–199. doi:[10.1016/j.jeconom.2020.09.006](https://doi.org/10.1016/j.jeconom.2020.09.006).
- Wing, C., Freedman, S.M., Hollingsworth, A., 2024. Stacked Difference-in-Differences. Working Paper 32054. National Bureau of Economic Research.

Online Supplementary Material

The Causal Effects of Korea's Quota Tariff Policy on Agricultural Product Retail Prices

S1 Summary Statistics

This section summarizes the estimation sample and variables in four tables; it also records the coding of the policy covariates of section 3.6 of the main text and the reasons for excluding three products from the sample. Table S1 reports descriptive statistics for the variables entering the LP-DiD regressions, while table S2 and table S3 report the annual imports of the 41 products and table S4 their pre-treatment import dependence. Table S1 is organized in three panels. Panel A covers the daily panel variables: the seasonally adjusted retail price from which the dependent variable is built, the daily Quota Tariff treatment status, and the covariates for the applied tariff rate, climate, oil prices, and WTO Market Access Quotas. For the two variables in panel B, summary statistics are computed not at the daily level but at the unit at which each variable is measured. Treatment intensity G , the tariff reduction induced by a designation, varies across ever-treated products but not over time,¹ and its summary statistics are therefore computed at the product level. Government stockpile releases are recorded as monthly totals in the raw data, which places their summary statistics at the product-month level. Panel C reports the seasonally adjusted retail price of each of the 41 products in four blocks: ever-treated leafy and root vegetables, ever-treated fruits, never-treated

¹ Treatment intensity G —the difference between the tariff rate that would apply in the absence of the designation (the applied tariff rate) and the rate under the designation—is in fact time-varying, since the applied tariff rate changes monthly. However, by construction the treatment-intensity specification in section 4.3 of the main text requires a time-invariant treatment intensity, and because G in this study is defined as the tariff reduction at the time the designation takes effect, we hold G fixed at that value.

vegetables and other crops, and never-treated fruits. The reported statistics (N, mean, standard deviation, minimum, and maximum) are the standard set in the empirical pass-through literature.

All entries in table S1 are level values in their raw units, prior to any transformation for use in the LP-DiD regressions. In the estimation itself, the dependent variable is the log of the seasonally adjusted retail price differenced against its average over the 120 days preceding the event, as defined in equation (1) of the main text, and the covariates enter in the forms defined in its sections 3.6 and 4.3. Reporting raw levels keeps the scale and range of each variable visible rather than hidden behind those transformations.

The policy covariates of section 3.6 of the main text are coded as follows. The Market Access Quota runs by calendar year, with cumulative imports counted from January 1; an expansion takes effect on its enactment date and remains in force through December 31, and a later expansion in the same year supersedes it from its own effective date. The one exception is ginger, whose 2023 expansion lapsed on September 30, 2023, by its own terms. The three covariates of the block Q_{it} are defined on every product-day, zero rather than missing, leaving no observation to drop from the regressions. The coverage dummy is included because the two kinds of zero are not equivalent: for an item outside the WTO TRQ system a zero expansion is forced, there being no quota to enlarge, whereas for a covered item it records the government's choice not to deploy an available instrument. The stockpile release records are monthly totals in tons; mapped onto the daily panel, each monthly total applies to every day of its month, and products absent from the records are coded zero. A release presupposes a stockpile, and a recorded zero therefore means different things for different products: for most items no stockpile program exists and the release value is necessarily zero, whereas for a stockpiled item a zero records a month in which the program chose not to release. The incidence dummy of the block R_{it} separates whether releases were under way from how large they were. Over the sample period the quota expansions for never-treated products fell on garlic, bean, mung bean, sesame, peanut and ginger, and the three stockpiled ever-treated products began their longest designation episodes in months with stockpile releases of 3,527 tons (onion), 955 tons (napa cabbage) and 1,959 tons (radish).

Table S1: Summary Statistics of Regression Variables

	N	Mean	SD	Min	Max
<i>Panel A. Daily panel variables</i>					
Retail price, seasonally adjusted (KRW/kg)	63,013	9,435.9	7,494.7	753.5	36,797.0
Quota Tariff treatment status (0/1)	63,591	0.062	0.240	0.000	1.000
Applied tariff rate (%)	63,591	35.5	42.5	0.0	587.9
Mean temperature (°C)	63,591	13.3	10.8	-14.9	31.8
Mean relative humidity (%)	63,591	65.2	13.8	22.9	99.3
Precipitation (mm)	63,591	3.8	13.6	0.0	176.2
Sunshine duration (hours)	63,345	6.2	4.0	0.0	13.5
Real crude oil price divided by the KRW/USD rate ($\times 1,000$)	63,591	20.7	3.9	14.2	34.9
WTO Market Access Quota coverage (0/1)	63,591	0.341	0.474	0.000	1.000
WTO quota expansion (ratio)	63,591	0.276	1.420	0.000	9.548
WTO quota expansion $\times$ tariff gap	63,591	1.441	8.258	0.000	56.334
<i>Panel B. Treatment intensity and policy variables</i>					
Treatment intensity G (%p; ever-treated items)	11	26.2	17.3	1.5	69.9
Government stockpile releases (tons/month; item $\times$ month)	2,091	70.0	388.9	0.0	5,214.0
<i>Panel C. Seasonally adjusted retail price (KRW/kg), by product</i>					
Ever-treated: leafy and root vegetables					
cabbage	1,548	2,180.8	331.4	1,504.5	3,061.4
carrot	1,548	4,060.7	820.0	2,703.9	6,072.3
green onion	1,548	3,130.0	825.9	1,711.7	6,361.5
napa cabbage	1,548	2,253.6	404.8	1,496.8	3,758.7
onion	1,548	2,146.6	363.4	1,505.0	3,088.1
radish	1,548	1,222.6	250.4	753.5	1,974.4
Ever-treated: fruits					
avocado	1,093	8,359.5	988.9	6,385.9	11,981.9
banana	1,548	2,851.3	300.3	1,956.4	3,346.3
kiwifruit	1,548	4,714.2	323.4	3,877.7	5,558.7
mango	1,548	13,505.1	2,057.3	8,666.4	19,104.1
pineapple	1,548	4,625.3	342.0	3,663.3	5,500.4
Never-treated: vegetables and other crops					
adzuki bean	1,548	16,149.5	2,725.4	12,131.5	24,999.6
bean	1,548	9,937.2	1,107.9	7,743.0	12,224.3
bell pepper	1,548	8,176.9	781.8	6,014.7	11,121.2
cherry tomato	1,548	8,345.8	836.9	6,072.9	10,715.3
cucumber	1,548	4,071.2	555.6	2,817.7	5,941.3
dried red pepper	1,548	26,393.4	3,793.7	20,620.9	36,797.0
garlic	1,548	10,297.2	1,983.8	6,741.8	13,866.9
ginger	1,548	11,304.9	2,363.3	7,155.3	16,390.4
green pepper	1,548	11,641.5	1,826.2	7,999.8	18,859.5
lettuce	1,548	11,134.9	1,917.4	7,419.2	18,435.2
mung bean	1,548	20,505.6	1,211.9	18,071.3	23,180.6
peanut	1,548	27,732.3	2,102.0	24,094.2	33,784.4
perilla leaf	1,548	23,060.0	2,337.6	17,495.5	29,706.7
red pepper	1,548	18,386.9	2,860.9	11,551.6	29,229.2
scallion	1,548	7,672.0	1,171.0	5,059.3	10,891.7

	N	Mean	SD	Min	Max
sesame	1,548	31,001.2	2,340.7	26,306.4	36,493.6
soybean sprout	1,548	4,572.9	487.7	3,136.4	5,585.0
spinach	1,548	9,404.9	1,544.8	5,566.8	15,590.7
squash	1,548	5,410.5	892.2	3,612.6	9,152.9
sweet pepper	1,548	12,274.1	1,731.0	8,625.9	18,976.1
sweet potato	1,548	5,159.0	716.9	3,927.9	6,941.1
tomato	1,548	5,516.2	682.9	3,919.8	7,626.1
water dropwort	1,548	11,250.7	3,194.3	6,435.8	17,156.8
young napa cabbage	1,548	2,941.1	405.6	2,063.8	4,214.5
young summer radish	1,548	3,220.3	726.4	1,956.0	6,398.6
Never-treated: fruits					
apple	1,548	8,779.5	1,114.7	6,599.0	12,383.1
lemon	1,548	6,747.0	457.0	5,584.8	7,976.3
melon	1,548	6,352.3	584.9	5,112.6	7,785.4
pear	1,548	5,875.6	1,180.7	3,200.4	9,583.2
watermelon	1,548	4,192.1	351.5	3,211.7	4,899.9

The estimation panel covers 41 products from January 1, 2021, to March 31, 2025. The retail-price series is seasonally adjusted (section S5); the panel is unbalanced because avocado’s series begins on April 4, 2022, lowering its N . Treatment status is the raw daily Quota Tariff indicator, whereas the treatment path used in estimation bridges gaps of at most 30 days (section S6). Treatment intensity G is a single time-invariant value per ever-treated product ($N = 11$), whose per-product values appear in table 2 of the main text; government stockpile releases are summarized at the product–month level. The oil variable, shown multiplied by 1,000 for readability, is the West Texas Intermediate crude oil price deflated by the U.S. consumer price index and divided by the KRW/USD exchange rate. In the regressions the climate variables enter as $[t - 100, t]$ averages and their one-year lags, and stockpile releases enter through an inverse-hyperbolic-sine transform with an incidence dummy (section 3.6 of the main text); the estimation sample varies by horizon under the clean-treated and clean-control conditions, and horizon-specific treated and control counts accompany the estimates.

Table S2 reports the annual import volume of the 41 products in metric tons, obtained by aggregating the monthly import records in the customs data to product–year totals. The rows follow the same four blocks and ordering as panel C of table S1. The salient contrast in the two import tables lies between the two groups of ever-treated products. The five ever-treated fruits show substantial imports in every year of the sample period (banana, for example, at roughly 400,000 tons in 2024), whereas the ever-treated vegetables, apart from carrot and onion, are imported on a far smaller scale and much more erratically. The never-treated products range from items with no recorded imports at all to heavily imported items such as bean. Section 5.3 of the main text relates the difference between the fruit and vegetable results to the share of each product’s supply that enters on the tariff lines covered by a designation. The retail prices of the four ever-treated tropical fruits are prices of the imports whose tariff a designation

reduces, and those of the ever-treated vegetables are prices of both domestic and imported produce, although none of their surveyed varieties is recorded as imported (table S13). Like import dependence, this share differs between the two product groups, so the estimates in the main text cannot test whether it causes the difference in results.

Table S2: Annual Import Volume by Product

	2021	2022	2023	2024	2025 (Jan.–Mar.)
Ever-treated: leafy and root vegetables					
cabbage	4,450.1	4,333.7	6,741.6	23,527.3	19,989.6
carrot	91,719.9	96,243.6	110,396.1	117,695.2	29,917.4
green onion	8,272.3	1,135.5	7,748.9	23,388.6	3,924.9
napa cabbage	66.7	2,030.1	164.2	4,135.2	5,383.4
onion	38,513.4	70,650.9	117,075.1	78,625.2	42,678.2
radish	7,693.8	2,461.1	1,307.3	14,973.3	8,700.2
Ever-treated: fruits					
avocado	17,313.1	13,479.2	12,671.2	11,251.6	2,450.1
banana	351,903.1	319,852.1	329,027.5	400,771.5	95,363.4
kiwifruit	41,000.8	44,350.8	37,191.8	49,649.9	2,280.4
mango	22,291.5	23,197.8	26,962.7	35,382.2	11,517.3
pineapple	61,688.1	67,104.1	67,047.7	81,573.9	16,720.9
Never-treated: vegetables and other crops					
adzuki bean	18,523.7	20,587.5	21,653.7	23,153.4	5,288.6
bean	248,122.2	285,928.0	247,294.5	263,993.9	78,519.5
bell pepper	0.0	4.4	0.0	0	5.8
cherry tomato	0.0	0	0	0	0
cucumber	0.0	0	0	0.0	0
dried red pepper	3,809.0	3,940.4	6,458.9	5,184.7	1,012.0
garlic	1,022.1	1,428.9	330.2	425.0	122.0
ginger	5,640.7	2,799.4	4,081.2	4,462.9	1,358.0
green pepper	2.0	0.0	11.1	5.5	6.4
lettuce	6,676.2	3,365.9	5,043.2	11,080.7	2,849.1
mung bean	16,213.3	12,193.8	14,001.7	14,067.9	3,424.5
peanut	30,955.4	31,372.5	30,695.9	33,307.8	10,278.6
perilla leaf	10,458.1	0	0.0	0	0
red pepper	2.0	0.0	11.1	5.5	6.4
scallion	2,320.8	22.7	38.2	116.3	34.7
sesame	86,922.9	84,250.2	71,932.7	82,216.7	18,829.3
soybean sprout	10,458.1	9,125.0	9,515.5	11,170.9	3,376.8
spinach	0	0.0	0.1	0.1	0
squash	26,193.6	19,808.5	12,315.7	22,312.8	7,906.1
sweet pepper	0.0	4.4	0.0	0	5.8
sweet potato	0.0	0	0	0.0	0
tomato	0.0	0	0	0	0
water dropwort	10,458.1	9,125.0	9,515.5	11,170.9	3,376.8
young napa cabbage	66.7	2,030.1	164.2	4,135.2	5,383.4
young summer radish	22,918.0	22,383.3	23,229.9	21,597.4	4,603.6
Never-treated: fruits					
apple	0.0	0	0	0.0	0

	2021	2022	2023	2024	2025 (Jan.–Mar.)
lemon	17,348.3	20,466.7	19,710.4	28,037.2	6,624.2
melon	1,944.6	1,727.1	3,150.3	3,434.6	0.0
pear	18.5	12.8	0.0	10.4	13.6
watermelon	0	0	0	0	0

Volume in metric tons. “0” denotes no recorded imports, whereas “0.0” denotes positive imports below 50 kg. The 2025 column covers January–March only and is not comparable to full calendar years. Products that share a ten-digit tariff line cannot be separated in the customs records and therefore report identical series: napa cabbage and young napa cabbage; bell pepper and sweet pepper; green pepper and red pepper; water dropwort and soybean sprout; tomato and cherry tomato.

Arranged in the same blocks, table S3 reports the corresponding annual import values in thousands of USD, complementing the volume figures of table S2. Because unit values differ widely across products, the two tables do not rank products alike: kiwifruit, for example, has smaller import volumes than pineapple yet substantially larger import values in each year of 2021–2024.

Table S3: Annual Import Value by Product

	2021	2022	2023	2024	2025 (Jan.–Mar.)
Ever-treated: leafy and root vegetables					
cabbage	1,607	1,485	1,953	8,542	6,083
carrot	42,528	44,096	38,877	51,065	12,831
green onion	8,952	1,120	4,496	15,163	2,216
napa cabbage	45	1,140	77	2,290	1,824
onion	17,111	32,760	46,869	25,788	12,827
radish	1,521	673	435	3,227	3,079
Ever-treated: fruits					
avocado	50,386	42,240	32,267	36,885	9,466
banana	290,151	283,876	304,746	382,043	77,219
kiwifruit	145,453	158,945	131,100	175,129	6,615
mango	95,276	102,212	112,206	161,171	44,174
pineapple	52,640	62,218	65,872	85,603	14,941
Never-treated: vegetables and other crops					
adzuki bean	37,175	43,531	34,537	32,895	7,759
bean	177,774	242,714	223,923	218,119	58,080
bell pepper	0	11	0	0	27
cherry tomato	0	0	0	0	0
cucumber	0	0	0	0	0
dried red pepper	10,654	11,846	23,566	16,418	2,876
garlic	1,836	2,342	636	934	257
ginger	7,915	1,378	6,900	5,628	1,592
green pepper	11	0	11	5	6
lettuce	5,337	2,967	4,969	9,738	2,113
mung bean	32,556	25,108	30,958	31,326	7,568
peanut	60,489	62,024	64,076	65,165	18,558
perilla leaf	18,476	0	0	0	0

	2021	2022	2023	2024	2025 (Jan.–Mar.)
red pepper	11	0	11	5	6
scallion	2,953	8	14	47	14
sesame	151,900	168,277	162,324	176,110	39,384
soybean sprout	18,476	18,562	18,456	16,898	3,715
spinach	0	0	1	1	0
squash	17,898	14,606	13,982	12,566	5,279
sweet pepper	0	11	0	0	27
sweet potato	0	0	0	0	0
tomato	0	0	0	0	0
water dropwort	18,476	18,562	18,456	16,898	3,715
young napa cabbage	45	1,140	77	2,290	1,824
young summer radish	12,474	12,729	12,144	16,834	4,146
Never-treated: fruits					
apple	1	0	0	0	0
lemon	38,678	43,260	42,271	58,915	13,167
melon	2,920	3,149	5,152	5,752	0
pear	90	71	0	49	56
watermelon	0	0	0	0	0

Value in thousands of USD, rounded to the nearest thousand; “0” denotes no recorded imports or a positive value below USD 500. The 2025 column covers January–March only and is not comparable to full calendar years. Products that share a ten-digit tariff line cannot be separated in the customs records and therefore report identical series: napa cabbage and young napa cabbage; bell pepper and sweet pepper; green pepper and red pepper; water dropwort and soybean sprout; tomato and cherry tomato.

Arranged in the same blocks, table S4 reports the pre-treatment import dependence of the 41 products, with block means and standardized differences between the ever-treated vegetables or fruits and the never-treated products.

Table S4: Pre-Treatment Import Dependence of Ever-Treated and Never-Treated Products

	Main 2019–2021 (%)	Reference 2021 (%)
Ever-treated: leafy and root vegetables		
cabbage	6.3	1.1
carrot	54.7	45.0
green onion	2.8 ^a	1.6 ^b
napa cabbage	21.4	0.0
onion	4.9	2.4
radish	1.2	0.7
<i>mean (number of products)</i>	15.2 (6)	8.5 (6)
Ever-treated: fruits		
avocado	100.0 ^c	100.0 ^c
banana	100.0 ^c	100.0 ^c
kiwifruit	63.9	67.3
mango	100.0 ^c	100.0 ^c
pineapple	100.0 ^c	100.0 ^c

	Main 2019–2021 (%)	Reference 2021 (%)
<i>mean (number of products)</i>	92.8 (5)	93.5 (5)
Never-treated: vegetables and other crops		
adzuki bean	72.9	78.7
bean	93.4	69.1
bell pepper	–	0.0
cherry tomato	16.3 ^a	–
cucumber	6.1	0.0
dried red pepper	82.1	3.9
garlic	12.0	0.3
ginger	35.2	13.6
green pepper	–	0.0
lettuce	–	6.4
mung bean	76.9 ^d	89.0
peanut	76.5 ^d	74.4
perilla leaf	–	–
red pepper	–	–
scallion	2.8 ^a	–
sesame	88.9	89.6
soybean sprout	–	–
spinach	–	0.0
squash	–	9.7
sweet pepper	–	0.0
sweet potato	0.0	0.0
tomato	16.3	0.0
water dropwort	–	–
young napa cabbage	–	–
young summer radish	–	–
<i>mean (number of products)</i>	44.6 (13)	25.6 (17)
Never-treated: fruits		
apple	0.0	0.0
lemon	100.0 ^c	100.0 ^c
melon	4.5 ^e	4.5
pear	0.0	0.0
watermelon	0.0 ^e	0.0
<i>mean (number of products)</i>	20.9 (5)	20.9 (5)
Never-treated, all: mean (number of products)	38.0 (18)	24.5 (22)
Standardized difference: vegetables	–0.72	–0.55
Standardized difference: fruits	1.82	2.42

Import dependence in percent, rounded to one decimal place. The main column divides imports by production plus imports minus exports, each summed over 2019–2021 (*Food Balance Sheet 2023*); the reference column divides the 2021 imports on each product’s own HSK lines by production plus imports. A dash in the main column means that the *Food Balance Sheet* has no production row or no trade row for the product; a dash in the reference column means that production is not reported for the product alone or that its HSK line also covers other products. Such missing values are filled by the assumptions in notes b, c, and e, except for the never-treated vegetables and other crops, whose missing values are left blank. Means are taken over products with a value, including the filled values, and the standardized differences compare each ever-treated group with all never-treated products, vegetables and fruits combined; the standardized difference is the difference in means divided by the square root of the average of the two

sample variances and is undefined when a group contains a single product. The text following the table describes the sources and the construction of both columns. ^a Green onion and scallion are both assigned the row for Welsh onion, and cherry tomato the row for tomato; these rows are not split, so each pair takes the same value in the main column. ^b Production is not reported for green onion alone; the reference value divides its 2021 imports by the 2021 production of the Welsh onion row, which also covers scallion, plus those imports. ^c No domestic production is recorded for banana, pineapple, mango, avocado, and lemon; their production is set to zero, so their import dependence is 100% in both columns. ^d The 2020–2021 import and export figures for mung bean and peanut, printed on each other’s lines, are reassigned on the basis of the supply figures in table 4.3; as printed, the ratios would be 92.9% for mung bean and 59.7% for peanut. ^e The *Food Balance Sheet* has no trade row for melon and watermelon; the main column takes the value of the reference column.

In table S4, import dependence is imports divided by domestic availability, measured in two ways. The main column sums each quantity over 2019–2021 before taking the ratio, with domestic availability equal to production plus imports minus exports; all three years precede the first Quota Tariff onset among the ever-treated products, on July 20, 2022. These quantities, in metric tons, come from tables 4.1 and 4.2 of *Food Balance Sheet 2023*, published by the Korea Rural Economic Institute. The import figures of table 4.2 appear to include processed products converted into raw-product equivalents, although the publication does not state its conversion rules. The import dependence of napa cabbage (21.4%) therefore includes kimchi imports, which account for almost all of it: in 2021, imports on the fresh napa cabbage line were 66.7 metric tons, 0.014% of the 462,031 metric tons of napa cabbage imports in the *Food Balance Sheet*. Because most napa cabbage is used to make kimchi, napa cabbage imports and kimchi imports serve the same market. The reference column divides the 2021 imports (the 2021 volumes in table S2) on each product’s own ten-digit lines of the Harmonized Tariff Schedule of Korea (HSK) by production plus imports, without subtracting exports. Subtracting the 2021 exports reported in the *Food Balance Sheet* changes this ratio by at most 0.90 percentage points for 18 of the 19 products with both a reference value and export data (e.g., cabbage: from 1.14% to 1.16%); the exception is dried red pepper (from 3.94% to 8.82%). Besides napa cabbage (21.4% against 0.0% in the reference column), the two columns differ widely for dried red pepper, bean, ginger, tomato, and garlic, presumably because the *Food Balance Sheet* also covers forms of these products outside the tariff lines used here, such as processed and dried goods.

Apart from the assumptions in notes c and e to table S4, a missing row in the *Food Balance Sheet* is not treated as zero, and trade in vegetables without their own row,

which may be recorded under the residual row for other vegetables, is not allocated to individual products because their shares of that row are unknown. Blank cells in table 4.2 (imports and exports of sweet potato; exports of bean, adzuki bean, and mung bean) are set to zero, since the annual balance sheets (table 3.11) report these flows as 0.0 thousand metric tons. The *Food Balance Sheet* reports the previous year's harvest as each year's production of fall-harvested crops such as sweet potato; for bean, adzuki bean, mung bean, peanut, sesame, and sweet potato, the columns of table 4.1 are used as printed, whereas the reference column uses the 2021 harvest.

Potato, grape and rice are excluded from the sample (section 3.1 of the main text) for the following reasons. The Quota Tariff on potato applies only to potatoes for chip manufacturing, whereas retail prices are observed only for table potatoes; the treatment and the dependent variable would therefore refer to different goods. Grape was designated once in May 2024, but its daily retail price series is unobserved from June 2023 onward, leaving no dependent variable over its event window. Rice, the staple grain, is managed under the government's priority food policy, under which its import and supply decisions differ from those of ordinary agricultural products.

S2 Quota Exhaustion Dates for Pineapple

This section works through the exhaustion calculation for each of the six pineapple designations, using Methods A and B as defined in section 3.2 of the main text. In table S5, Start date and End date delimit the designated period announced in advance by the government, irrespective of any subsequent quota exhaustion, and Quota (ton) is the quota announced alongside it. For No. 1, Method A shows cumulative imports falling short of the quota through the End date, whereas Method B places the crossing on December 21, 2022. Since Method B overstates volumes, the authoritative Method A record prevails, and pineapple is taken never to have exhausted its quota during the No. 1 period. For No. 2, Method A establishes that the quota was exceeded at some point in October 2023, because no imports cleared under the Quota Tariff are recorded for November or December; Method B dates the crossing to October 9, 2023, and this date is adopted. Reflecting this, the cells from October 9, 2023, onward within the No. 2 period are white in table 1 of the main text (not-treated status).

Table S5: Quota Tariff Designations for Pineapple

No.	Start date	End date	Quota (ton)
1	Nov. 10, 2022	Dec. 31, 2022	8600
2	Aug. 25, 2023	Dec. 31, 2023	5000
3	Jan. 19, 2024	June 30, 2024	40000 → unlimited
4	July 1, 2024	Sept. 30, 2024	unlimited
5	Oct. 1, 2024	Dec. 31, 2024	unlimited
6	Jan. 24, 2025	June 30, 2025	46000

For No. 3, Method A records imports cleared under the Quota Tariff all the way to the End date. Method B, computed against the initially announced quota of 40,000 tons, places the crossing of cumulative total imports on June 29, 2024, only one day before the End date. Moreover, that quota had been revised to unlimited on April 5, 2024. Under either method, therefore, pineapple never exhausted its quota during No. 3, and the entire designated period appears as black cells in table 1 of the main text (treated status). For No. 4 and No. 5, the government announced the quota as unlimited from the outset, leaving no exhaustion calculation to perform, and both periods are black cells throughout. Finally, No. 6 shows no exceedance under either method through March 31, 2025, the end of the sample period, and is likewise black throughout that span.

S3 Tariff Rate Priority Structure

To determine the final applicable rate among multiple tariff rates that may apply to a single HSK code, this study follows the priority order of rate application set out in *Article 50 of the Customs Act*. The order has seven levels, described below.

The rates applied with the highest priority are Special Tariff Rates designed to protect domestic industries or correct trade imbalances, including Anti-Dumping Duty, Retaliatory Duty, and Countervailing Duty. The second-priority rate is the FTA rate under the *Act on Special Cases of the Customs Act for the Implementation of Free Trade Agreements*. If this rate is equal to or lower than rates under the *Customs Act* (excluding first-priority rates), the FTA rate is applied preferentially upon the importer's request. Third-priority rates include International Cooperation Tariffs, such as WTO General

Tariff Concessions, and Beneficial Tariffs. Among third-priority rates, Tariff Concessions for agricultural, forestry, and livestock products (conceded at rates equivalent to the domestic-foreign price differential or at rates exceeding Basic Tariff Rates in conjunction with domestic market opening under the *Regulations on Tariff Concessions in the Framework of the Agreement Establishing the World Trade Organization*) take precedence over Basic Tariff Rates and Provisional Tariff Rates. Fourth-priority rates include Adjusted Duty², Quota Tariff, and Seasonal Tariffs. The Quota Tariff takes precedence when it is lower than Preferential Tariffs for least developed countries, and it also supersedes Basic Tariff Rates and Provisional Tariff Rates. The fifth-priority rate is Preferential Tariffs for least developed countries. Provisional Tariff Rates and Basic Tariff Rates correspond to the sixth- and seventh-priority rates, respectively. The applied tariff rate of section 3.4 of the main text has one further limitation. Where the law prescribes charging whichever of the ad valorem duty and the Specific Duty is larger in amount, the duty actually charged cannot be determined, because the price and weight of individual import transactions are unobservable, and the applied tariff rate is therefore computed from ad valorem rates only.

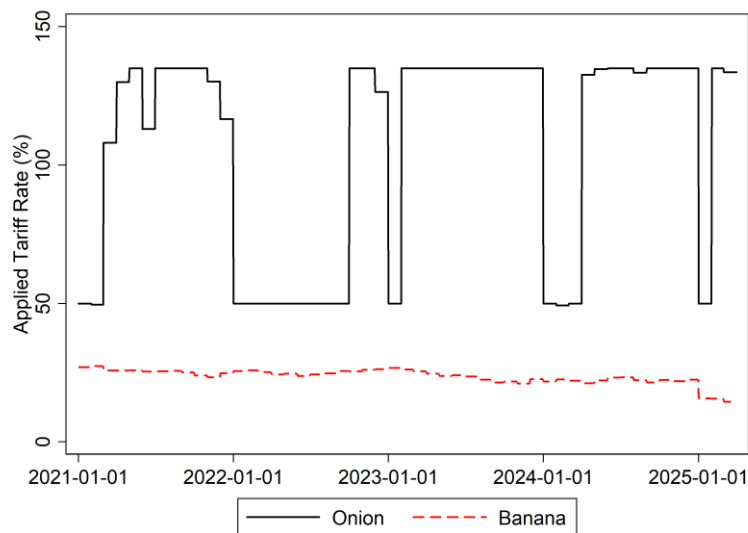
S4 Deriving the Applied Tariff Rate: Worked Examples

As of July 2024, the import shares for banana under HSK code 0803.90-0000 were highest for the Philippines, followed by Vietnam and Ecuador. Because the applied tariff rate is the tariff that would prevail absent the Quota Tariff, the Quota Tariff then in force for banana is left out of the calculation. Ecuadorian banana is not subject to first-priority rates (such as Anti-Dumping Duty, Retaliatory Duty, or Countervailing Duty), and Ecuador is not an FTA partner country; therefore, the review begins with the third-priority rates. Since the fourth- through sixth-priority rates other than the Quota Tariff do not apply and the third-priority WTO General Tariff Concession rate (90%) exceeds the seventh-priority Basic Tariff Rate (30%), the Basic Tariff Rate is ultimately applied.

² Article 69, Subparagraphs 1, 3, and 4 of the Customs Act

Vietnamese banana is subject to a 0% tariff under the Korea-Vietnam FTA, while Philippine banana, which accounts for approximately 61% of imports, is subject to the 30% Basic Tariff Rate. Weighting these country-specific rates by import volume shares yields an applied tariff rate of approximately 23.3% for banana in July 2024. With the Korea-Philippines FTA entering into force on December 31, 2024, the applied tariff rate for banana declined to approximately 14% to 15% as of the first quarter of 2025, as illustrated in figure S1.

Figure S1: Applied Tariff Rate (Banana, Onion)



Fresh unpeeled onion is not subject to first-priority rates; thus, the review begins with the second-priority rates. Chinese onion accounts for the majority of total import volume, but it derives no practical benefit from the FTA Tariff Concessions, and therefore the third-priority rates apply. The third-priority rate is the WTO Tariff Concession applicable to agricultural, forestry, and livestock products with an established Market Access Quota. Imports within the Market Access Quota are subject to the Recommended Tariff Rate of 50%, while imports exceeding the quota are subject to the non-recommended rate of 135%. Consequently, the monthly applied tariff rate for onion varies depending on whether the Market Access Quota is exceeded. This explains why the applied tariff rate for onion in figure S1 varies substantially between 50% and 135%. Values that deviate from exactly 50% or 135% reflect onion imports originating from countries other than China.

S5 Removing Seasonality

This paper removes seasonality via a regression-based adjustment employing calendar dummy variables and Fourier terms (Pierce et al., 1984). For daily data, we include day-of-week dummies, model annual seasonality using Fourier (sine/cosine) terms, and control for major and moving holidays with dedicated dummies; the seasonally adjusted series is obtained by subtracting the estimated seasonal components from the raw data.

Two considerations motivated this choice. The first is the need to keep policy effects out of the seasonal component: Quota Tariff designations often coincide with periods of seasonal price fluctuation, and the adjustment must therefore distinguish price movements attributable to the policy from recurring seasonal patterns. A natural candidate for daily deseasonalization is STL, a non-parametric method that extracts seasonality at any specified period and has been applied extensively to weekly and daily data (Cleveland et al., 1990).³ Since STL relies on non-parametric smoothing, however, trends or policy shocks in the data may be partially absorbed during the decomposition, and treatment effects removed at the seasonal-adjustment stage cannot be recovered in the subsequent DiD analysis.⁴

The regression-based adjustment eliminates this risk. The calendar dummies and Fourier terms enter as deterministic regressors, independent of the treatment variable, which lets the seasonal component be estimated separately rather than confounded with trends or policy effects. No seasonal pattern remains in the adjusted series, while trend changes and the effects of Quota Tariff designations are fully preserved.

The second consideration is the structure of the raw series itself. The daily data exhibit strong, multi-layered seasonality: week-of-year and day-of-week patterns, holiday effects, and an annual cycle that is not even fixed in length, some years comprising 52 weeks and others 53. Standard official seasonal-adjustment tools such as X-12-ARIMA/X-13ARIMA-SEATS are not directly applicable to such data, because

³ Multiple STL (MSTL) extends the method to extract several seasonal components simultaneously, such as 7-day and annual cycles (Bandara et al., 2025).

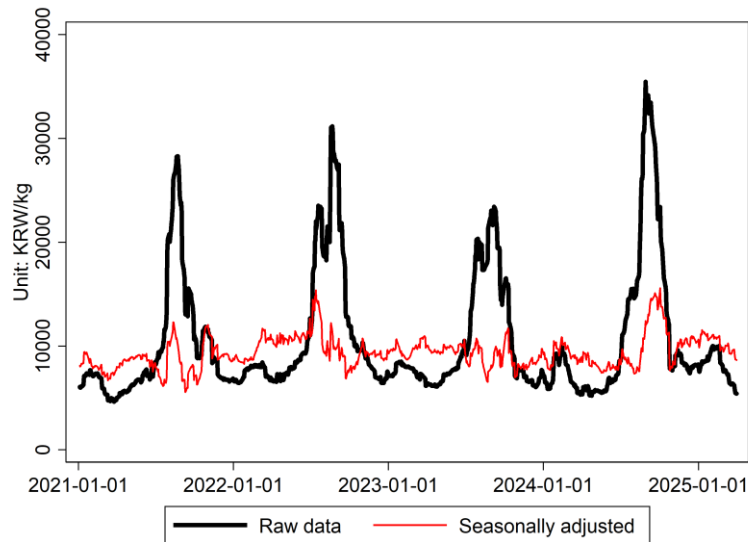
⁴ STL-based methods also require the seasonal period to be a fixed whole number of observations, so the extra day in leap years has to be handled outside the procedure: Cleveland et al. (1990) delete every February 29 to keep the period at 365 days. Nor does the standard procedure include regressors for moving holidays: applied to weekly unemployment insurance claims in the United States, MSTL leaves large spikes at the end of each year, which Mollins and Lumb (2024) attribute to improper holiday treatment.

they are designed for monthly or quarterly series and assume fixed 12-month or 4-quarter cycles (Mollins and Lumb, 2024).

The regression-based approach is established practice for exactly this situation: the U.S. Bureau of Labor Statistics (BLS), when seasonally adjusting weekly unemployment insurance claims, developed and has employed MoveReg, a moving weighted regression based on weekly dummies and Fourier terms, instead of traditional X-13 techniques (Mollins and Lumb, 2024).

Figure S2 shows the result for spinach: the adjusted series retains the level and trend of the original while the recurring within-year swings are removed.

Figure S2: Seasonally Adjusted Retail Price (Spinach)



S6 Additional Details of the Estimation Design

Clean Treatment Events

Table S6 lists every treatment onset among the ever-treated products after bridging gaps of at most 30 days. For each onset, the table reports the spell length, the gap since the previous episode, and the clean/dirty classification implied by the 120-day washout condition. Three onsets are dirty: carrot (October 29, 2024; 52-day gap), pineapple

(January 19, 2024; 102-day gap), and cabbage (January 24, 2025; 103-day gap). All three are therefore excluded from the treated side in every specification, leaving nine vegetable and eight fruit events.

Table S6: Quota Tariff Onsets and Washout Classification

Group	Product	Onset date	Spell (days)	Gap (days)	Classification
1	Cabbage	2024-05-10	156	—	clean
1	Cabbage	2025-01-24	67 [†]	103	dirty (washout)
1	Carrot	2024-05-10	120	—	clean
1	Carrot	2024-10-29	154 [†]	52	dirty (washout)
1	Green onion	2022-07-20	104	—	clean
1	Green onion	2023-05-01	61	181	clean
1	Green onion	2023-11-17	166	139	clean
1	Napa cabbage	2024-05-10	326 [†]	—	clean
1	Onion	2022-08-17	177	—	clean
1	Radish	2023-05-01	61	—	clean
1	Radish	2024-07-01	274 [†]	366	clean
2	Avocado	2024-01-19	438 [†]	—	clean
2	Banana	2022-11-10	52	—	clean
2	Banana	2023-11-17	501 [†]	320	clean
2	Kiwifruit	2024-04-05	271	—	clean
2	Mango	2022-11-10	52	—	clean
2	Mango	2023-08-25	585 [†]	236	clean
2	Pineapple	2022-11-10	52	—	clean
2	Pineapple	2023-08-25	45	236	clean
2	Pineapple	2024-01-19	438 [†]	102	dirty (washout)

[†] Spell right-censored at the end of the sample period (March 31, 2025).

Spells are measured after bridging gaps of at most 30 days; gaps are days since the previous episode ended.

Treated-Set Attrition in the Persistent-Treatment Specification

Under the stay-on condition, an event remains in the horizon- h estimation sample only while h is smaller than its spell length, so the spell lengths in table S6 determine exactly which events remain in the treated set at each horizon. The persistent-treatment LP-DiD graphs in the main text are truncated at $h = 150$ because, on exactly the day of the compositional change at $h = 156$ (the end of the cabbage spell), the coefficient of the vegetable treatment-status specification rises abruptly from +0.4 to +18.2 log points. This discontinuity is not a genuine effect of the Quota Tariff treatment but the result of a break in the sample composition caused by events failing the stay-on condition. The

sharp rise in the vegetable coefficients beyond $h = 156$ must therefore not be interpreted as a causal effect of the Quota Tariff. In the impulse-response specification, there is no stay-on condition in the first place; no event ever exits the sample, and the composition of the treated set is identical at every horizon.

The Washout Window and Long-Horizon Tracking

The washout condition treats a Quota Tariff onset as a clean treated event when the product has never been treated during the preceding $L = 120$ days, under the assumption that the dynamic effects of a previously treated episode stabilize within 120 days (Dube et al., 2025). The impulse-response specification, however, tracks each onset for up to 273 days from the day it begins, so a later onset that satisfies the washout condition can still fall within the tracking window of an earlier one. This happens when the earlier episode ended at least 120 days before the later onset but began no more than 273 days before it. The later onset is then classified as treated, and the earlier onset also remains classified as treated, because the impulse-response specification requires no stay-on condition in the first place. As a result, the two onsets are counted as treated events at the same time, and both enter the sample of the LP-DiD regressions. In this study’s data, this overlap occurs once: the green onion onset of November 17, 2023, falls 200 days after that of May 1, 2023. Raising L to 140 days would remove this overlap at the cost of discarding the November 2023 onset; this paper retains $L = 120$.

S7 Joint Tests of the Pre-Onset Coefficients

Table S7 reports, for the treatment-status and treatment-intensity specifications in each group, the three joint tests of the 120 pre-onset coefficients discussed in section 4.4 of the main text. A Wald test that all 120 coefficients are zero cannot be computed: with standard errors clustered by product, the estimated covariance matrix of the coefficients has rank at most the number of product clusters minus one, so the 120×120 matrix is singular.

Table S7: Joint Tests of the Pre-Onset Coefficients, $h \in [-120, -1]$

	Vegetables		Fruits	
	Status	Intensity	Status	Intensity
Product clusters	36	36	35	35
<i>Sup-t simultaneous 95% band</i>				
max $ t $	2.42	1.83	2.39	2.80
Critical value	2.90	2.91	3.03	3.06
Leads outside the band	0	0	0	0
<i>p</i> -value	0.157	0.459	0.228	0.100
<i>Linear pre-trend (per 30 days)</i>				
Slope	3.65	0.071	0.59 [†]	0.022 [†]
95% CI	[-1.74, 9.05]	[-0.080, 0.222]	[-1.19, 2.37] [†]	[-0.052, 0.096] [†]
<i>p</i> -value	0.178	0.348	0.507 [†]	0.548 [†]
<i>Four 30-day windows</i>				
<i>F</i> test <i>p</i> -value	0.266	0.762	0.304	0.467

Coefficients and slopes in log points; treatment-intensity columns are per 1%p of tariff reduction. The vegetable regressions include the climate covariates and the fruit regressions do not, as in the main text. Product-clustered standard errors. The sup- t critical value is simulated from the estimated joint covariance of the 120 coefficients (100,000 draws); the slope test uses the t distribution and the window test the F distribution, both with (denominator) degrees of freedom equal to the number of product clusters minus one. [†] Estimated on the observations whose outcomes are available over all 120 pre-onset days.

S8 Removing the Ever-Treated Product with the Largest Tariff Reduction

In the treatment-intensity specification, β^h is expressed per 1%p of tariff reduction and is a weighted combination of the effects of the individual onsets, each per 1%p of its own reduction (section 4.3 of the main text). If the effect of an onset is not proportional to G_i , the product with the largest tariff reduction could weigh heavily in β^h . Table S8 therefore re-estimates the six treatment-intensity regressions of table 3 of the main text after removing, in each group, the ever-treated product with the largest tariff reduction: onion among the vegetables and pineapple among the fruits. The signs and the 5% significance of all six estimates are unchanged. For fruits, the persistent-treatment estimate at $h = 150$ is almost identical, because both clean pineapple onsets leave the sample before that horizon (spells of 52 and 45 days in table S6) and the removal drops only pineapple's control observations from before its first onset. The price responses

after the pineapple onsets are weak relative to their tariff reductions and pull the fruit impulse-response estimates toward zero, so these estimates become larger in absolute value without pineapple.

Table S8: Treatment-Intensity Estimates without the Ever-Treated Product with the Largest Tariff Reduction

Group (product removed)	Specification	h	Without the product	Table 3	Treated products (onsets)
Vegetables (onion)	Persistent treatment	150	-0.143 (0.785)	-0.117 (0.822)	4 (4)
	Impulse response	150	-0.286 (0.529)	-0.268 (0.553)	5 (8)
	Impulse response	250	1.379 (<0.001)	1.379 (<0.001)	5 (8)
Fruits (pineapple)	Persistent treatment	150	-0.787 (0.001)	-0.786 (0.001)	4 (4)
	Impulse response	150	-0.390 (0.053)	-0.088 (0.725)	4 (6)
	Impulse response	250	-0.769 (0.015)	-0.495 (0.044)	4 (6)

Coefficients $\times 100$, in log points per 1%p of tariff reduction, from the treatment-intensity specifications; p -values in parentheses, based on product-clustered standard errors and the t distribution with degrees of freedom equal to the number of product clusters minus one. The column “Table 3” repeats the corresponding treatment-intensity estimates of table 3 of the main text. The last column counts the treated products and onsets that remain after the removal. Onion is the ever-treated vegetable with the largest tariff reduction ($G_i = 69.9$, against 27 to 29 for the other ever-treated vegetables), and pineapple the ever-treated fruit with the largest ($G_i = 29.6$, against at most 25 for the other ever-treated fruits). No pineapple onset remains in the persistent-treatment regression at $h = 150$ (table S6), so that cell is not informative as a check.

S9 Summary of the Robustness Checks

Table S9 summarizes how the treatment-intensity estimates of table 3 of the main text change under the washout, fixed-set, spillover and comparison-pool checks of sections 5.1 to 5.3 of the main text, and also reports the pre-trend tests and the check that drops the product with the largest tariff reduction.

Table S9: Summary of Robustness Checks

	Vegetables		Fruits		Same
	$h = 150$	$h = 250$	$h = 150$	$h = 250$	
Table 3 of the main text	−0.117 (0.822)	1.379 (<0.001)	−0.786 (0.001)	−0.495 (0.044)	
180-day washout	−0.111 (0.830)	1.379 (<0.001)	−0.786 (0.001)	−0.495 (0.044)	Yes
240-day washout	−0.111 (0.830)	1.379 (<0.001)	−0.819 (n.a.)	−0.302 (0.256)	No
First onsets only	−0.214 (n.a.)	1.032 (0.038)	0.891 (n.a.)	−0.108 (0.652)	No
Fixed set of treated onsets	−0.117 (0.822)	−	−0.786 (0.001)	−	Yes
Substitutes and complements removed	−0.280 (0.656)	0.730 (0.010)	−0.829 (<0.001)	−0.544 (0.036)	Yes
Import-dependent comparison pool	−	−	−1.050 (<0.001)	−0.720 (0.032)	Yes
Largest-intensity product dropped	−0.143 (0.785)	1.379 (<0.001)	−0.787 (0.001)	−0.769 (0.015)	Yes
Pre-trend tests (p)	0.46 / 0.35 / 0.76		0.10 / 0.55 / 0.47		

The $h = 150$ columns are persistent-treatment estimates and the $h = 250$ columns impulse-response estimates, both from the treatment-intensity specifications of table 3 of the main text, in log points per 1%p of tariff reduction; p -values in parentheses use product-clustered standard errors and the t distribution with degrees of freedom equal to the number of product clusters minus one; n.a.: three or fewer treated products; −: not estimated. “Same” is Yes when every entry in the row keeps the sign and the 5% significance of table 3. The substitutes and complements removed are candidates selected from estimated cross-price elasticities; the import-dependent comparison pool is used for fruits only (12 product clusters). The fixed set consists of the onsets that remain at $h = 150$, so its entries equal those of table 3; with the largest-intensity product dropped, the fruit $h = 150$ entry is almost unchanged because no pineapple onset remains at that horizon. The last row reports, for the same treatment-intensity specifications, the p -values of the three joint tests of the pre-onset coefficients over $h \in [-120, -1]$ in section 4.4 of the main text (simultaneous band, linear trend and four 30-day windows). Details are in tables S7, S8, S10 to S12 and S16.

S10 The Washout Condition

This section reports the checks of section 5.1 of the main text. Table S10 re-estimates the six regressions of table 3 of the main text for each group with washout periods of 120 days (the main specification), 180 days and 240 days, and on the first onset of each product only.

Table S10: Estimates under Longer Washout Periods and with First Onsets Only

	120 days	Washout period 180 days	240 days	First onset only
<i>Group 1 (Vegetables)</i>				
Treatment status				
Persistent, $h = 150$	-3.894 (0.793)	-3.738 (0.802)	-3.738 (0.802)	-6.555 (n.a.)
Initiation, $h = 150$	-7.770 (0.547)	-7.502 (0.563)	-7.423 (0.572)	-10.708 (0.534)
Initiation, $h = 250$	38.813 (<0.001)	38.813 (<0.001)	38.813 (<0.001)	29.318 (0.028)
Treatment intensity				
Persistent, $h = 150$	-0.117 (0.822)	-0.111 (0.830)	-0.111 (0.830)	-0.214 (n.a.)
Initiation, $h = 150$	-0.268 (0.553)	-0.259 (0.568)	-0.256 (0.577)	-0.379 (0.538)
Initiation, $h = 250$	1.379 (<0.001)	1.379 (<0.001)	1.379 (<0.001)	1.032 (0.038)
Treated products / onsets				
Persistent, $h = 150$	5 / 5	4 / 4	4 / 4	3 / 3
Initiation, $h = 250$	6 / 9	6 / 8	6 / 7	6 / 6
<i>Group 2 (Fruits)</i>				
Treatment status				
Persistent, $h = 150$	-8.950 (0.156)	-8.950 (0.156)	-8.297 (n.a.)	-0.101 (n.a.)
Initiation, $h = 150$	-2.338 (0.599)	-2.338 (0.599)	-1.966 (0.732)	2.727 (0.489)
Initiation, $h = 250$	-11.519 (0.049)	-11.519 (0.049)	-6.895 (0.264)	-4.011 (0.533)
Treatment intensity				
Persistent, $h = 150$	-0.786 (0.001)	-0.786 (0.001)	-0.819 (n.a.)	0.891 (n.a.)
Initiation, $h = 150$	-0.088 (0.725)	-0.088 (0.725)	-0.110 (0.730)	0.206 (0.290)
Initiation, $h = 250$	-0.495 (0.044)	-0.495 (0.044)	-0.302 (0.256)	-0.108 (0.652)
Treated products / onsets				
Persistent, $h = 150$	4 / 4	4 / 4	3 / 3	2 / 2
Initiation, $h = 250$	5 / 8	5 / 8	5 / 6	5 / 5

Coefficients $\times 100$: treatment status in log points, treatment intensity in log points per 1%p of tariff reduction; p -values in parentheses, based on product-clustered standard errors and the t distribution with degrees of freedom equal to the number of product clusters minus one. n.a.: three or fewer treated products remain, too few for inference. The column “120 days” reproduces table 3 of the main text. Only the treated-side washout changes; the 120-day pre-onset baseline and the control-side condition are unchanged. With a 365-day washout the estimates coincide with those at 240 days for vegetables and with the first-onset estimates for fruits, so no 365-day column is shown.

Table S11 addresses the change in the treated set across horizons in the persistent-treatment specification. Panel A keeps, at every horizon from $h = 0$ to $h = 150$, only the onsets that remained under the Quota Tariff for at least the first 150 days: onion (August 17, 2022), green onion (November 17, 2023), napa cabbage and cabbage (May 10, 2024) and radish (July 1, 2024) among the vegetables, and mango (August 25, 2023), banana (November 17, 2023), avocado (January 19, 2024) and kiwifruit (April 5, 2024) among the fruits. Only the treated side is fixed; the control side follows the rules of the baseline design, which panel B reproduces, and at $h = 150$ both designs coincide with table 3 of the main text. For fruits the two designs coincide from $h = 60$,

and at the shorter horizons estimated ($h = 0, 10, \dots, 50$, of which the table shows $h = 0$ and $h = 30$; the other estimates are in the replication files) no fruit estimate is significant at the 5% level in either design. For vegetables they coincide from $h = 120$, and the positive coefficients in the first weeks after adoption appear in both designs.

Table S11: Persistent-Treatment Estimates with a Fixed Set of Treated Onsets

	Vegetables		Fruits	
	Status	Intensity	Status	Intensity
<i>Panel A. Fixed set of treated onsets</i>				
$h = 0$	1.731 (0.781)	−0.031 (0.803)	0.427 (0.843)	0.119 (0.213)
$h = 30$	17.580 (0.003)	0.334 (0.166)	−2.996 (0.250)	−0.008 (0.947)
$h = 60$	20.743 (0.062)	0.713 (0.062)	0.138 (0.964)	0.103 (0.567)
$h = 90$	1.270 (0.913)	−0.061 (0.895)	−4.782 (0.028)	−0.346 (0.001)
$h = 120$	5.662 (0.595)	0.152 (0.687)	−5.573 (0.116)	−0.437 (0.010)
$h = 150$	−3.894 (0.793)	−0.117 (0.822)	−8.950 (0.156)	−0.786 (0.001)
Treated onsets (every h)	5	5	4	4
<i>Panel B. All clean onsets remaining at each horizon (baseline design)</i>				
$h = 0$	2.838 (0.537)	0.023 (0.856)	2.193 (0.247)	0.123 (0.173)
$h = 30$	15.345 (0.003)	0.368 (0.067)	−1.634 (0.429)	0.008 (0.917)
$h = 60$	13.953 (0.160)	0.494 (0.141)	0.138 (0.964)	0.103 (0.567)
$h = 90$	7.683 (0.464)	0.191 (0.628)	−4.782 (0.028)	−0.346 (0.001)
$h = 120$	5.662 (0.595)	0.152 (0.687)	−5.573 (0.116)	−0.437 (0.010)
$h = 150$	−3.894 (0.793)	−0.117 (0.822)	−8.950 (0.156)	−0.786 (0.001)
Treated onsets ($h = 0 / h = 150$)	9 / 5	9 / 5	8 / 4	8 / 4

Coefficients $\times 100$: treatment status in log points, treatment intensity in log points per 1%p of tariff reduction; p -values in parentheses, based on product-clustered standard errors and the t distribution with degrees of freedom equal to the number of product clusters minus one. Panel A keeps, at every horizon, the onsets that remained under the Quota Tariff for at least the first 150 days; panel B is the baseline design of table 3 and the figures of the main text. Only the treated side differs between the panels.

S11 The Stable Unit Treatment Value Assumption

The sensitivity analysis for spillovers in section 5.2 of the main text removes candidate substitutes and complements of the ever-treated products from the comparison pool. The candidates are selected from compensated cross-price elasticities estimated with the almost ideal demand system (AIDS) of Deaton and Muellbauer (1980) for 40 products. The data are the weekly point-of-sale sales revenue and average prices of each product at large retailers and supermarket chains, published by *Nongnet* and compiled from

Marketlink data, for the 153 weeks from ISO week 14 of 2022 to ISO week 10 of 2025; quantities are sales revenue divided by the average price. Each product's series is seasonally adjusted with multiplicative ISO-week factors and interpolated to the 1,065 days from April 7, 2022, to March 6, 2025, by piecewise cubic Hermite interpolation. The system is estimated jointly, with symmetry, homogeneity and adding-up imposed (858 free demand parameters), by nonlinear least squares (NLS) followed by feasible generalized NLS (FGNLS) iterated to convergence. The compensated (Hicksian) elasticities are evaluated at mean prices and mean expenditure. Because the independent information in the data consists of the 153 weekly observations, the delta-method standard errors of the elasticities are multiplied by $\sqrt{1,065/153} \approx 2.64$. The code for the demand system, the selection of candidates and the exclusion specifications is included in the replication files. The estimated elasticities are also shown on an interactive map at the [repository link anonymized for review]. For the 24 pairs of products that the authors judge to be substitutes or complements, the estimated compensated elasticity has the expected sign in 18; the list of pairs and the comparison are included in the replication files.

Let e_{ck} denote the compensated elasticity of the quantity demanded of product c with respect to the price of ever-treated product k . For each k , the exclusion list contains the products c for which e_{ck} has a Benjamini–Hochberg q -value (Benjamini and Hochberg, 1995) of at most 0.01 in the converged estimates and passes a further check: a q -value of at most 0.10, with the same sign as in the converged estimates, when the NLS point estimates are combined with panel-corrected standard errors (Beck and Katz, 1995), which allow for correlation and differing variances across products. The exclusion lists thus contain only the most clearly estimated relationships. Each list is split by sign: products with a positive e_{ck} form the substitute list of k and those with a negative e_{ck} its complement list; by the symmetry of compensated effects, e_{ck} and e_{kc} have the same sign. Lists are built for all 11 ever-treated products. For the never-treated products, the multiple-testing correction is applied over all cells that pair a never-treated product with an ever-treated product. Because ever-treated products also serve as controls before their first onset, the candidates for each list also include the other ever-treated products of the same group. For these, the correction is applied separately, over the 50 cells that pair two different ever-treated products of the same group; cells that pair an ever-treated vegetable with an ever-treated fruit are not used, since the two

groups are estimated in separate regressions. Of these 50 cells, only 19 can enter a comparison, because e_{ck} matters only when an onset of k precedes the first onset of c ; repeating the correction over these 19 cells alone leaves the classification of all 19 unchanged.

Three exclusion specifications are estimated. The one reported in section 5.2 of the main text removes both substitutes and complements; the other two remove only substitutes or only complements. In each, a control observation of product c on comparison date t , the onset date of the treated events it is compared with, is dropped from the horizon- h regression ($h \geq 0$) if an ever-treated product of either group whose list contains c has the Quota Tariff in force on any day of $[t - 120, t + h]$, that is, from the first day of the 120-day pre-onset baseline through the date at which the outcome is measured. The condition is added to the control side of equations (3) and (6) of the main text; apart from it and the removal of the three products described below, the specifications of table 3 of the main text are unchanged. Because the calendar-day fixed effects restrict every comparison to observations on the same date, the rule defines the comparison pool separately for each comparison date and horizon. When the Quota Tariff is in force for several ever-treated products within the window, as after the concurrent designations of napa cabbage, cabbage and carrot on May 10, 2024, their lists are combined. Young napa cabbage, scallion and young summer radish, closely related to napa cabbage, green onion and radish, are dropped from every vegetable comparison in all three exclusion specifications, whatever their estimated elasticities. None of these removals applies to table 3 or the figures of the main text.

Table S12 reports the three exclusion specifications beside the estimates of table 3 of the main text. Over $h \in [0, 150]$ in the persistent-treatment specification and $h \in [0, 250]$ in the impulse-response specification, at least 21 control products remain on every comparison date in both groups with substitutes and complements removed, at least 24 with only substitutes removed and at least 23 with only complements removed. With both removed, all 12 estimates of table 3 of the main text keep their signs and 5% significance. The main fruit estimate (treatment intensity, persistent treatment, $h = 150$) changes from -0.786 to -0.829 log points per 1%p ($p < 0.001$). At $h = 250$ in the impulse-response specification, the fruit declines become slightly larger, from -11.519 to -12.483 log points for treatment status ($p = 0.042$) and from -0.495 to

−0.544 log points per 1%p for treatment intensity ($p = 0.036$). All four vegetable estimates at $h = 150$ remain insignificant, and the positive vegetable coefficients at $h = 250$ in the impulse-response specification fall from 38.813 to 21.250 for treatment status and from 1.379 to 0.730 for treatment intensity, while remaining significant ($p = 0.010$ in both cases). The partial variants are reported separately because substitute and complement spillovers bias the estimates in opposite directions, and the specification that removes both cannot show which of them was present. With only substitutes removed or only complements removed, all 12 estimates also keep their signs and 5% significance.

The selection of candidates from the demand system has limitations. The demand system treats prices as exogenous and estimates 858 demand parameters from 153 weeks of data on the 40 products. On 77.6% of the 1,065 days used to estimate it, at least one ever-treated product had the Quota Tariff in force under the treatment definition of the main text, so the estimated elasticities may also reflect the supply shocks that prompted designations. The classification into substitutes and complements is sensitive to how the standard errors are estimated, and some estimated relationships are hard to explain economically. The exclusion lists therefore contain candidates selected by the estimated elasticities, and they do not establish which products are substitutes or complements. For the same reason, the three products named above are removed from every vegetable comparison in every exclusion specification, whatever their estimated elasticities.

Table S12: Estimates with Candidate Substitutes and Complements Removed from the Comparison Pool

	(1) Quota Tariff Treatment Status Group 1 (Vegetables)	(2) Treatment Status Group 2 (Fruits)	(3) Quota Tariff Treatment Intensity Group 1 (Vegetables)	(4) Treatment Intensity Group 2 (Fruits)
<i>Panel A: Persistent treatment, $h = 150$</i>				
Table 3	−3.894 (0.793)	−8.950 (0.156)	−0.117 (0.822)	−0.786 (0.001)
Substitutes and complements removed	−8.162 (0.650)	−9.225 (0.157)	−0.280 (0.656)	−0.829 (<0.001)
Only substitutes removed	−7.141 (0.672)	−9.153 (0.161)	−0.241 (0.685)	−0.830 (<0.001)
Only complements removed	−6.017 (0.699)	−8.840 (0.166)	−0.198 (0.714)	−0.780 (0.002)
<i>Panel B: Initiation (impulse response), $h = 150$</i>				
Table 3	−7.770 (0.547)	−2.338 (0.599)	−0.268 (0.553)	−0.088 (0.725)
Substitutes and complements removed	−15.804 (0.229)	−2.920 (0.518)	−0.562 (0.212)	−0.125 (0.622)
Only substitutes removed	−9.817 (0.490)	−2.909 (0.515)	−0.345 (0.489)	−0.126 (0.616)
Only complements removed	−13.775 (0.257)	−2.233 (0.620)	−0.486 (0.245)	−0.083 (0.740)
<i>Panel C: Initiation (impulse response), $h = 250$</i>				
Table 3	38.813 (<0.001)	−11.519 (0.049)	1.379 (<0.001)	−0.495 (0.044)
Substitutes and complements removed	21.250 (0.010)	−12.483 (0.042)	0.730 (0.010)	−0.544 (0.036)
Only substitutes removed	39.810 (<0.001)	−12.238 (0.044)	1.387 (<0.001)	−0.543 (0.033)
Only complements removed	21.424 (<0.001)	−11.785 (0.048)	0.752 (<0.001)	−0.500 (0.046)
<i>Minimum number of control products on a comparison date</i>				
Substitutes and complements removed	21	23	21	23
Only substitutes removed	24	26	24	26
Only complements removed	23	26	23	26

Coefficients $\times 100$: treatment status in log points, treatment intensity in log points per 1%p of tariff reduction; p -values in parentheses, based on product-clustered standard errors and the t distribution with degrees of freedom equal to the number of product clusters minus one (33 product clusters in the vegetable regressions after the three products named below are removed, and 35 in the fruit regressions). The rows “Table 3” repeat the estimates of table 3 of the main text. In each exclusion specification, a control observation of product c on comparison date t , the date of an onset that enters the regression, is dropped from the horizon- h regression if an ever-treated product whose exclusion list contains c has the Quota Tariff in force on any day of $[t - 120, t + h]$; the lists contain candidate substitutes, candidate complements or both, selected by compensated cross-price elasticities from the demand system described in this section. Young napa cabbage, scallion and young summer radish are dropped from every vegetable comparison in all three exclusion specifications. The last block of rows reports the smallest number of control products remaining on any comparison date over $h \in [0, 150]$ in the persistent-treatment specification and $h \in [0, 250]$ in the impulse-response specification; within each group it is the same for treatment status and treatment intensity.

S12 The Difference between Fruit and Vegetable Estimates

The tables in this section report the evidence behind section 5.3 of the main text, and table S15 also reports its checks of the fruit estimates for specifications and horizons that the main text does not discuss; the code that produces them is included in the replication files. Table S13 lists the varieties measured by the retail price series of the ever-treated vegetables and tropical fruits, table S14 reports policy exposure and the price change it predicts from pre-designation information, table S15 checks the fruit estimates against alternative explanations, table S16 re-estimates them with the never-treated part of the comparison pool restricted to import-dependent products, and table S17 tests the difference between the fruit and vegetable estimates directly.

The minimum detectable effect for carrot reported in section 5.3 of the main text is computed for a one-sided 5% test with 80% power; its range depends on how the standard deviation of a single-onset estimate is measured.

The forgone tariff revenue reported in section 5.3 of the main text is the customs value of the imports on the tariff lines for fresh produce within each designated period, up to the quota volume where one was set, multiplied by the difference between the applied tariff rate and the Quota Tariff rate, assuming that the same quantities would have been imported without the designations; the calculation is included in the replication files. The designations of frozen and dried green onion and of prepared pineapple, which fell within designated periods of the fresh products, are not counted, because the estimated price effects, the treatment intensity and policy exposure all refer to the tariff lines for fresh produce. For the vegetables, the range depends on whether the tariff that onion imports would otherwise have faced is capped at the 50% in-quota rate of its Market Access Quota or set at the applied tariff rate, which also reflects the higher over-quota rate.

Table S13: Varieties Measured by the Retail Price Series of the Ever-Treated Vegetables and Tropical Fruits

Product	Series used	Varieties surveyed (days observed)	Imported variety
Vegetables			
napa cabbage	all varieties	autumn (158), highland (130), spring (222), highland summer (159), overwintered (421)	no
radish	all varieties	autumn (139), highland (282), spring (177), overwintered (528)	no
cabbage	all varieties	cabbage (1,041)	no
green onion	green onion	green onion (1,041)	no
onion	all varieties	onion (1,037), new-crop onion (30)	no
carrot	unwashed	unwashed (1,041)	no
Fruits			
avocado	all varieties	imported (733)	yes
mango	all varieties	imported (897)	yes
banana	all varieties	imported (1,041)	yes
pineapple	all varieties	imported (1,041)	yes
Retail price data: January 4, 2021, to March 31, 2025			

Varieties surveyed in the retail price data used as the dependent variable, with the number of days on which each variety is observed. “Imported variety” indicates whether any surveyed variety is recorded as imported; a “no” does not mean that the series excludes imported produce, because the retail price of each product includes imported produce.

Table S14: Policy Exposure and the Price Change Predicted from Pre-Designation Information

Product	Clean onsets	Policy exposure s	Quota Tariff rate (%)	Tariff reduction (%p)	Predicted change (log points)
Vegetables					
napa cabbage ^a	1	0.000	0	27.0	0.00
radish	2	0.007	0	28.7 to 30.0	-0.17
cabbage	1	0.013	0	27.0	-0.31
green onion	3	0.018	0	27.0	-0.43
onion	1	0.026	10	69.9	-1.28
carrot	1	0.495	0	28.7	-12.48
Fruits					
kiwifruit	1	0.659	5	1.5	-0.91
avocado	1	1.000	0	8.0	-7.68
mango	2	1.000	0	15.5 to 15.8	-14.64 to -14.44
banana	2	1.000	0	24.2 to 24.9	-22.25 to -21.65
pineapple	2	1.000	0	29.5 to 30.0	-26.21 to -25.84

Policy exposure s is 2021 customs imports on the tariff lines for fresh produce later covered by a designation divided by domestic supply, defined as average 2019–2021 production in the *Food Balance Sheet* plus those imports; exports are not subtracted, and the four tropical fruits, which have no domestic production data, have $s = 1$. Under full pass-through with proportional margins, the predicted change is $100 \times s \times \ln[(1 + \tau)/(1 + \tau + \Delta)]$, where τ is the Quota Tariff rate and Δ the tariff reduction; it is computed without using prices, and apart from the tariff rates, its only inputs are customs import records for the tariff lines for fresh produce and the *Food Balance Sheet* of the Korea Rural Economic Institute. Tariff reductions are those at each clean onset and may differ from the product averages over all onsets in table 2 of the main text; ranges cover the clean onsets of each product. ^a In 2021, imports on the fresh napa cabbage line were 66.7 metric tons, 0.014% of the 462,031 metric tons of napa cabbage imports in the *Food Balance Sheet*, which are almost entirely kimchi (table S4); no kimchi tariff line was designated between 2015 and 2025.

Panels A and B of table S15 check the fruit estimates against alternative explanations. When product fixed effects and product-specific coefficients on the level of the KRW/USD exchange rate are added to the specification of table 3 of the main text, the treatment-intensity estimate under persistent treatment at $h = 150$ remains negative and significant ($p < 0.001$), whereas the two impulse-response estimates at $h = 250$, slightly larger in absolute value than in table 3, are not significant at the 5% level ($p = 0.129$ and $p = 0.132$). With product-specific linear time trends, the fruit estimates in all four columns keep their signs and their significance classification at the 5% level. Assigning the fruit treatment paths to the five never-treated fruits, which checks for a decline common to fruit prices in general, does not reproduce the decline. Nor does shifting the paths 730 days earlier, to placebo dates, which checks for trends in the prices of the ever-treated fruits. Panel C reports the wholesale and margin results

discussed in section 5.3 of the main text.

Table S15: Checks of the Fruit Estimates against Alternative Explanations

	Persistent treatment, $h = 150$		Impulse response, $h = 250$	
	Status	Intensity	Status	Intensity
<i>Panel A. Retail price</i>				
Baseline (table 3)	−8.950 (0.156)	−0.786 (0.001)	−11.519 (0.049)	−0.495 (0.044)
Product fixed effects and product-specific slopes on the KRW/USD rate	−13.878 (0.088)	−1.226 (<0.001)	−11.804 (0.129)	−0.587 (0.132)
Product-specific linear time trends	−17.526 (0.077)	−1.527 (0.001)	−27.671 (0.027)	−1.344 (0.029)
<i>Panel B. Fruit treatment paths assigned to other products or dates (retail price)</i>				
Assigned to the five never-treated fruits	4.450 (0.722)	−	2.392 (0.750)	−
Shifted 730 days earlier	−0.883 (n.a.)	−	6.169 (0.150)	−
<i>Panel C. Other outcomes</i>				
Wholesale price	−14.073 (0.056)	−1.092 (0.002)	−9.058 (0.185)	−0.374 (0.234)
Retail-to-wholesale margin	4.457 (0.102)	0.285 (0.118)	−3.038 (0.528)	−0.135 (0.558)

Coefficients $\times 100$: treatment status in log points, treatment intensity in log points per 1%p of tariff reduction; p -values in parentheses, based on product-clustered standard errors and the t distribution with degrees of freedom equal to the number of product clusters minus one. The exchange-rate row adds product fixed effects and product-specific coefficients on the level of the KRW/USD exchange rate to the specification of table 3 of the main text, and the linear-trend row adds only product-specific linear time trends to the same specification. The main specification has no product fixed effects because the outcome in equation (1) of the main text subtracts each product's pre-event mean. Wholesale prices are national wholesale market prices obtained from *Nongnet* and seasonally adjusted in the same way as retail prices; the margin outcome is the log of the retail-to-wholesale price ratio, transformed as in equation (1) of the main text. The five never-treated fruits are lemon, melon, pear, apple and watermelon. In both rows of panel B, the assigned paths carry no tariff reduction, so the treatment-intensity cells are not defined (−). n.a.: too few treated products for inference.

Table S16 re-estimates the fruit regressions with a comparison pool restricted to import-dependent never-treated products, because table S4 shows that the full comparison pool is much less import-dependent than the ever-treated fruits. The restricted pool contains the never-treated products whose import dependence in table S4 is at least 50% (dried red pepper, mung bean, peanut, sesame, bean and adzuki bean) and lemon, whose value there is set to 100% because no domestic production is recorded; a second version lowers the threshold to 30%, which adds ginger. The check is not applied to vegetables, whose comparison pool is already more import-dependent than the ever-treated vegetables. The fruit regressions with the restricted pool have 12 product clusters (13 with ginger), and most of the never-treated products in the pool are

subject to state trading and WTO Market Access Quotas, which partly insulate their domestic prices from world prices.

Table S16: Fruit Estimates with Import-Dependent Comparison Products

	(1) Quota Tariff Treatment Status	(2) Quota Tariff Treatment Intensity
<i>Panel A: Persistent treatment, $h = 150$</i>		
Table 3	−8.950 (0.156)	−0.786 (0.001)
Import dependence 50% or more, plus lemon	−14.693 (0.018)	−1.050 (<0.001)
Import dependence 30% or more, plus lemon	−12.589 (0.074)	−0.998 (<0.001)
<i>Panel B: Initiation (impulse response), $h = 150$</i>		
Table 3	−2.338 (0.599)	−0.088 (0.725)
Import dependence 50% or more, plus lemon	−5.396 (0.265)	−0.164 (0.567)
Import dependence 30% or more, plus lemon	−6.738 (0.200)	−0.284 (0.367)
<i>Panel C: Initiation (impulse response), $h = 250$</i>		
Table 3	−11.519 (0.049)	−0.495 (0.044)
Import dependence 50% or more, plus lemon	−17.300 (0.021)	−0.720 (0.032)
Import dependence 30% or more, plus lemon	−19.400 (0.025)	−0.902 (0.025)
<i>Never-treated comparison products (product clusters)</i>		
Table 3	30 (35)	30 (35)
Import dependence 50% or more, plus lemon	7 (12)	7 (12)
Import dependence 30% or more, plus lemon	8 (13)	8 (13)

Coefficients $\times 100$: treatment status in log points, treatment intensity in log points per 1%p of tariff reduction; p -values in parentheses, based on product-clustered standard errors and the t distribution with degrees of freedom equal to the number of product clusters minus one. Import dependence is the main measure of table S4 (2019–2021). At the 50% threshold, the never-treated comparison products are dried red pepper, mung bean, peanut, sesame, bean and adzuki bean, plus lemon; at the 30% threshold, ginger is added. The pre-onset observations of the ever-treated fruits stay in the comparison pool, as in table 3 of the main text.

Table S17: Direct Test of the Difference between Fruit and Vegetable Estimates

Specification	h	Vegetables	Fruits	Difference (fruits – vegetables)	95% CI	p -value
<i>Treatment status</i>						
Persistent treatment	150	−3.894	−8.950	−5.056 [16.030]	[−37.454, 27.342]	0.754
Impulse response	150	−7.770	−2.338	5.432 [13.518]	[−21.888, 32.752]	0.690
Impulse response	250	38.813	−11.519	−50.331 [9.004]	[−68.530, −32.133]	<0.001
<i>Treatment intensity</i>						
Persistent treatment	150	−0.117	−0.786	−0.670 [0.570]	[−1.821, 0.482]	0.247
Impulse response	150	−0.268	−0.088	0.181 [0.512]	[−0.855, 1.216]	0.726
Impulse response	250	1.379	−0.495	−1.874 [0.343]	[−2.568, −1.180]	<0.001
Product clusters	41					

Coefficients $\times 100$: treatment status in log points, treatment intensity in log points per 1%p of tariff reduction; standard errors of the difference in brackets. Confidence intervals and p -values use the t distribution with degrees of freedom equal to the number of product clusters minus one.

S13 Alternatives for Inference with Few Treated Clusters

This section supplements section 5.4 of the main text. It describes the distortions of cluster-robust inference and of the restricted wild cluster bootstrap when the treated clusters are few, then works through the alternatives to conventional cluster-robust inference and states why each is unavailable in this setting.

Cluster-robust t -statistics tend to over-reject when the treated clusters are few, because these clusters contribute too few residual blocks for the variance of the treatment coefficient to be estimated reliably (Conley and Taber, 2011; MacKinnon and Webb, 2020). The restricted wild cluster bootstrap errs in the opposite direction: it under-rejects in exactly this configuration, and its most recent variants do so severely when the number of treated clusters is small (MacKinnon et al., 2023). MacKinnon and Webb (2018) conclude that when the restricted and unrestricted bootstrap p -values diverge, as they often do with few treated clusters, neither can be trusted. The alternatives reviewed below restore validity only by imposing further assumptions; Roth et al. (2023) note that “all of the available approaches require the researcher to impose some potentially strong additional assumptions.”

Randomization and placebo procedures (Conley and Taber, 2011; Hagemann, 2019;

MacKinnon and Webb, 2020) infer the distribution of the treated clusters’ shocks from the control clusters, which is valid only when the two are exchangeable, and MacKinnon and Webb (2020) show that, with few treated clusters, randomization inference over-rejects or under-rejects when the error variance of the treated clusters differs from that of the control clusters. Exchangeability is untenable here: designation itself selects products prone to price surges, and daily price volatility differs sharply between the vegetables and the fruits. Ferman and Pinto (2019) relax exchangeability, but their correction requires the heteroskedasticity to follow a known parametric function of an observable⁵ by which the control residuals can be rescaled to stand in for the treated clusters’ shocks. The present panel offers no such observable: it is balanced, each product–day observation is a single price rather than a cell average, and the volatility differences follow no known parametric structure. Canay et al. (2021) derive conditions under which the wild bootstrap remains valid with few clusters, but these require homogeneity of the covariate design across clusters and are, by the authors’ own account, difficult to satisfy when treatment is assigned at the cluster level. Even the recent extensions to staggered timing and heterogeneous effects (Alvarez and Ferman, 2023) maintain a binary absorbing treatment and a modeled variance structure.

References

- Alvarez, L., Ferman, B., 2023. Extensions for Inference in Difference-in-Differences with Few Treated Clusters. doi:[10.48550/arXiv.2302.03131](https://doi.org/10.48550/arXiv.2302.03131), [arXiv:2302.03131](https://arxiv.org/abs/2302.03131).
- Bandara, K., Hyndman, R.J., Bergmeir, C., 2025. MSTL: A seasonal-trend decomposition algorithm for time series with multiple seasonal patterns. *International Journal of Operational Research* 52, 79–98. doi:[10.1504/IJOR.2025.143957](https://doi.org/10.1504/IJOR.2025.143957).
- Beck, N., Katz, J.N., 1995. What to do (and not to do) with time-series cross-section data. *American Political Science Review* 89, 634–647. doi:[10.2307/2082979](https://doi.org/10.2307/2082979).
- Benjamini, Y., Hochberg, Y., 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 289–300. doi:[10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x).
- Canay, I.A., Santos, A., Shaikh, A.M., 2021. The Wild Bootstrap with a “Small” Number of “Large”

⁵ In their leading case, the number of underlying observations averaged into each group×time cell.

- Clusters. *The Review of Economics and Statistics* 103, 346–363. doi:[10.1162/rest_a_00887](https://doi.org/10.1162/rest_a_00887).
- Cleveland, R.B., Cleveland, W.S., McRae, J.E., Terpenning, I., 1990. STL: A seasonal-trend decomposition procedure based on loess. *Journal of Official Statistics* 6, 3–73.
- Conley, T.G., Taber, C.R., 2011. Inference with “Difference in Differences” with a Small Number of Policy Changes. *The Review of Economics and Statistics* 93, 113–125. doi:[10.1162/REST_a_00049](https://doi.org/10.1162/REST_a_00049).
- Deaton, A., Muellbauer, J., 1980. An Almost Ideal Demand System. *American Economic Review* 70, 312–326.
- Dube, A., Girardi, D., Jordà, Ò., Taylor, A.M., 2025. A Local Projections Approach to Difference-in-Differences. *Journal of Applied Econometrics* 40, 741–758. doi:[10.1002/jae.70000](https://doi.org/10.1002/jae.70000).
- Ferman, B., Pinto, C., 2019. Inference in Differences-in-Differences with Few Treated Groups and Heteroskedasticity. *The Review of Economics and Statistics* 101, 452–467. doi:[10.1162/rest_a_00759](https://doi.org/10.1162/rest_a_00759).
- Hagemann, A., 2019. Placebo inference on treatment effects when the number of clusters is small. *Journal of Econometrics* 213, 190–209. doi:[10.1016/j.jeconom.2019.04.011](https://doi.org/10.1016/j.jeconom.2019.04.011).
- MacKinnon, J.G., Nielsen, M.Ø., Webb, M.D., 2023. Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics* 232, 272–299. doi:[10.1016/j.jeconom.2022.04.001](https://doi.org/10.1016/j.jeconom.2022.04.001).
- MacKinnon, J.G., Webb, M.D., 2018. The wild bootstrap for few (treated) clusters. *The Econometrics Journal* 21, 114–135. doi:[10.1111/ectj.12107](https://doi.org/10.1111/ectj.12107).
- MacKinnon, J.G., Webb, M.D., 2020. Randomization inference for difference-in-differences with few treated clusters. *Journal of Econometrics* 218, 435–450. doi:[10.1016/j.jeconom.2020.04.024](https://doi.org/10.1016/j.jeconom.2020.04.024).
- Mollins, J., Lumb, R., 2024. Seasonal Adjustment of Weekly Data. Staff Discussion Paper 2024-17. Bank of Canada.
- Pierce, D.A., Grupe, M.R., Cleveland, W.P., 1984. Seasonal Adjustment of the Weekly Monetary Aggregates: A Model-Based Approach. *Journal of Business & Economic Statistics* 2, 260–270. doi:[10.1080/07350015.1984.10509393](https://doi.org/10.1080/07350015.1984.10509393).
- Roth, J., Sant’Anna, P.H.C., Bilinski, A., Poe, J., 2023. What’s trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics* 235, 2218–2244. doi:[10.1016/j.jeconom.2023.03.008](https://doi.org/10.1016/j.jeconom.2023.03.008).